First-Mover Bias in Gradient Boosting Explanations:

Mechanism, Detection, and Resolution

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Abstract

We investigate a specific form of explanation instability that we term *first-mover bias*—a path-dependent concentration of feature importance associated with sequential residual fitting in gradient boosting—as a mechanistic contributor to the well-known instability of SHAP-based feature rankings under multicollinearity. When correlated features compete for early splits, gradient boosting creates a self-reinforcing advantage for whichever feature is selected first: subsequent trees inherit modified residuals that favor the incumbent, concentrating SHAP importance on an arbitrary feature rather than distributing it across the correlated group. Scaling up a single model amplifies this effect—a Large Single Model with the same total tree count as our method produces the *poorest* attribution reproducibility of any approach tested.

We provide evidence that *model independence* is sufficient to largely neutralize first-mover bias in the linear, high-collinearity regime we study, and that it remains the most effective mitigation under nonlinear data-generating processes. Both our proposed method, DASH (Diversified Aggregation for Stable Hypotheses), and simple seed-averaging (Stochastic Retrain) restore stability by breaking the sequential dependency chain, supporting the view that the operative mechanism is independence between explained models, not any particular aggregation strategy. At $\rho = 0.9$, both methods achieve stability ≈ 0.977–0.978, while the standard single-best workflow degrades to 0.958 and the Large Single Model to 0.938. On the Breast Cancer dataset, DASH improves stability from 0.376 to 0.925 (+0.549) over Single Best, and from 0.339 to 0.925 (+0.586) over the training-budget-matched Single Best (M=200), outperforming even Stochastic Retrain by +0.063. Under nonlinear data-generating processes, the stability advantage emerges at $\rho \geq 0.7$; at lower correlation, simpler methods suffice.

DASH additionally provides two diagnostic tools—the Feature Stability Index (FSI) and Importance-Stability (IS) Plot—that detect first-mover bias without ground truth and can be applied independently of the full DASH pipeline, enabling practitioners to audit explanation reliability before acting on feature rankings. Software and reproducible benchmarks are available at https://github.com/DrakeCaraker/dash-shap.

Keywords: first-mover bias, SHAP, feature importance, multicollinearity, model independence, gradient boosting, explainability, Rashomon effect

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1 Introduction

The standard workflow for explaining gradient-boosted tree predictions is deceptively simple: train a model, compute SHAP values, report the feature importance ranking. This ranking drives consequential decisions across science, industry, and regulation—it selects features for production pipelines, generates hypotheses in biomedical research, and satisfies regulatory auditors reviewing algorithmic systems. The workflow is ubiquitous, but under correlated predictors its resulting feature rankings may be substantially less reliable than their routine use suggests. When features are correlated, SHAP-based importance rankings can become unstable. At each split, a gradient-boosted tree must choose one feature from a correlated set. Since correlated features carry nearly identical predictive signal, the choice is governed by marginal numerical differences—effectively arbitrary. The model's predictions are robust to this choice, but the SHAP values are not: changing the random seed, the learning rate, or the tree depth can swap the positions of correlated features in the importance ranking without meaningfully altering predictive accuracy. This is a manifestation of the Rashomon effect [Breiman, 2001] applied to explanations rather than predictions.

The problem is pervasive because multicollinearity is pervasive. In clinical datasets, radius, perimeter, and area measure the same underlying tumor geometry. In materials science, atomic-level properties of constituent elements are correlated by construction. In economics and social science, income, education, and occupation form tightly coupled clusters. Any dataset where features were not specifically decorrelated before modeling is susceptible.

Why bigger models make it worse. The intuitive response to unstable explanations is to train a more powerful model. Our results suggest that this can be counterproductive. A single large XGBoost model with thousands of trees and the same total tree count as DASH ($M \times \sim 75 \approx 15{,}000$ trees, where ~ 75 is the average number of trees per population model after early stopping) produces the poorest attribution reproducibility among all methods tested. We hypothesize, and our experiments support, that the primary mechanism is *sequential residual dependency*: in gradient boosting, each tree fits the residuals of all previous trees. If tree 1 selects feature A from a correlated pair (A, B), it partially removes A's signal from the residuals. Subsequent trees find both A and B less useful, but A slightly more so because the residual structure favors the feature that was partially captured. Over thousands of iterations, this creates a "first-mover advantage" that concentrates importance on whichever feature happened to be selected first—an artifact of optimization path dependence, not a property of the data. We use the term *first-mover bias* to denote this specific path-dependent concentration effect in sequential boosting; our claim is not that this is the only source of SHAP instability, but that it is an important and previously under-emphasized contributor in gradient-boosted trees under correlation. We note that with low `colsample_bytree` (0.1–0.5 in our population), correlated features A and B are not always present in the same column sample, so the first-mover advantage is *probabilistic rather than deterministic*: over many trees, whichever feature accumulates slightly more early selections gains a cumulative residual advantage that is attenuated—but not eliminated—by column subsampling.

Our contribution. We propose DASH (Diversified Aggregation for Stable Hypotheses), a five-stage pipeline that produces stable and reproducible feature importance explanations. DASH is not intended to outperform tuned single models on prediction; its objective is reproducible attribution under multicollinearity while maintaining competitive predictive performance. A substantial portion of explanation instability appears to arise from individual models' optimization paths rather than solely from the data distribution itself: training enough diverse models and averaging their SHAP

values causes path-dependent arbitrary choices to cancel, producing a more reproducible importance ranking. Specifically, DASH:

1. Trains a population ($M = 200$) of XGBoost models with randomly sampled hyperparameters, including deliberately low `colsample_bytree` (0.1–0.5) to force feature diversity.
2. Filters for predictive quality, retaining only models within ε of the best validation score.
3. Selects a diverse subset ($K \leq 30$) via greedy max-min dissimilarity on feature utilization vectors.
4. Computes interventional TreeSHAP for each selected model and averages the SHAP matrices element-wise.
5. Provides diagnostic tools (Feature Stability Index, IS Plot, local disagreement maps) for auditing explanation reliability.

DASH operates at two layers. The *core operation*—averaging attributions across independently trained models—is established in companion work as the minimum-variance unbiased estimator (Cramér–Rao bound) and is shown to be Pareto-optimal among all stable attribution methods [Caraker et al., 2026b]. This core operation requires only model independence; even simple seed averaging (Stochastic Retrain) implements it. The *pipeline* (Stages 1–5) adds diversity enforcement, performance filtering, and stability diagnostics that improve finite-sample equity and provide actionable audit output beyond what plain averaging offers.

We make five contributions:

- **Mechanistic framing.** Feature selection bias in ensemble methods is well known [Strobl et al., 2007, Hooker and Mentch, 2019]; we articulate first-mover bias as a specific path-dependent concentration effect arising from sequential residual dependency in gradient boosting, and provide evidence that it concentrates SHAP-based feature attributions on arbitrary features under collinearity.
- **Principle.** We provide evidence that model independence is sufficient to largely neutralize first-mover bias in the linear, high-collinearity regime we study, and that it remains the most effective mitigation under nonlinear data-generating processes. Both DASH (deliberate diversity) and Stochastic Retrain (seed diversity) restore stability to the same level, consistent with the view that the operative mechanism is independence rather than any particular aggregation strategy.
- **Diagnostics.** The Feature Stability Index (FSI) and Importance-Stability (IS) Plot detect first-mover bias without access to ground truth, enabling practitioners to audit explanation reliability.
- **Method.** DASH (Diversified Aggregation for Stable Hypotheses) is an engineered pipeline that operationalizes the independence principle with forced feature restriction, diversity-aware model selection, and integrated diagnostics.
- **Infrastructure.** The `fit_from_attributions()` interface decouples the DASH aggregation stage from XGBoost, making the pipeline applicable to any attribution method (LIME, Integrated Gradients, neural network SHAP) that produces feature-level attribution vectors. This positions DASH as a general-purpose stability layer for model explanation rather than an XGBoost-specific tool (validated on LIME attributions in Appendix G).

On synthetic data, our "accuracy" metric evaluates agreement with a predefined equitable decomposition within correlated groups; it should be interpreted as agreement with a chosen attribution target rather than as direct recovery of uniquely identifiable ground truth (Section 5).

2 Related Work

SHAP and model explanations. SHAP values [Lundberg and Lee, 2017] provide a theoretically grounded decomposition of predictions into feature contributions, drawing on the Shapley value framework from cooperative game theory [Shapley, 1953]. TreeSHAP [Lundberg et al., 2020] enables efficient exact computation for tree-based models. While SHAP satisfies desirable axiomatic properties (local accuracy, missingness, consistency), these guarantees are conditioned on a fixed model. When the model changes— even slightly—the SHAP values can change substantially.

Instability of feature importance. The instability of SHAP values under model perturbation has been noted by several authors. Fisher et al. [2019] formalize the Rashomon set—the collection of models with near-optimal performance—and show that variable importance can vary widely across this set. Dong and Rudin [2020] visualize the "cloud" of variable importance across Rashomon-set models, demonstrating that importance rankings are inherently ambiguous when many near-optimal models exist. Semenova et al. [2022] further characterize the Rashomon set's structure and its implications for model selection. Bilodeau et al. [2024] prove impossibility results for complete and linear attribution methods (including SHAP and Integrated Gradients), showing they can provably fail to improve on random guessing for inferring model behavior on sufficiently rich model classes. Chen et al. [2024] provide the first systematic analysis of the Rashomon effect in gradient boosting specifically, with an information-theoretic characterization of the Rashomon set and a framework for mitigating predictive multiplicity. Marx et al. [2023] demonstrate that SHAP-based feature attributions are sensitive to reference distribution choices. Covert et al. [2021] provide a unified framework connecting removal-based explanations (including SHAP) and discuss stability considerations across methods. The problem is especially acute for correlated features, where SHAP must distribute credit among features that carry overlapping information [Kumar et al., 2020, Chen et al., 2020].

Ensemble explanations. Paillard et al. [2025] argue for computing SHAP values from a single large ensemble rather than aggregating explanations from multiple models, on the grounds that a single ensemble provides a consistent explanation. Our results challenge this recommendation: the single-ensemble approach (our "Ensemble SHAP" baseline) does not resolve the instability caused by correlated features and, when taken to its extreme (the Large Single Model), amplifies it.

Explanation multiplicity. Hwang et al. [2026] introduce *explanation multiplicity*— the phenomenon of multiple internally valid but substantively different SHAP explanations for the same decision—and present methodology to disentangle sources due to model training/selection from stochasticity intrinsic to the explanation pipeline. They show that apparent stability depends on the metric: magnitude-based distances can remain near zero while rank-based measures reveal substantial churn in top-feature identity. Our work complements theirs by providing a mechanistic account of the model-training-induced component: first-mover bias explains *why* retraining gradient-boosted models produces the rank-level instability that Hwang et al. characterize. The FSI diagnostic detects this instability from a single model ensemble without requiring repeated runs. Decker et al. [2024] propose optimal convex combinations of feature attributions to improve robustness or faithfulness;

DASH takes a different approach, enforcing diversity at the model level rather than optimizing the aggregation of a fixed set of explanations.

Stable explanations. Alvarez-Melis and Jaakkola [2018] propose metrics for explanation stability. Krishna et al. [2022] study disagreement among different explanation methods applied to the same model. Our work addresses a different source of disagreement: the same explanation method (SHAP) applied to different-but-equally-valid models.

Explanation reliability and aggregation. Molnar et al. [2022] provide a general framework for model-agnostic feature importance that connects permutation-based and SHAP-based approaches, noting that instability under feature dependence is a shared concern across methods. Slack et al. [2020] demonstrate that SHAP explanations can be sensitive to adversarial perturbations of the classifier, raising broader concerns about explanation reliability beyond the multicollinearity setting we address.

Feature selection under multicollinearity. Classical approaches to multicollinearity include variance inflation factors [O'Brien, 2007], principal component regression, and elastic net regularization [Zou and Hastie, 2005]. Stability selection [Meinshausen and Bühlmann, 2010] addresses a related problem by subsampling data and tracking which features are consistently selected across subsamples. Permutation importance [Altmann et al., 2010] offers an alternative to SHAP with different stability properties. Causal Shapley values [Heskes et al., 2020] provide principled handling of correlated features through causal structure. These methods operate at the model-fitting or feature-selection stage. DASH operates at the explanation stage, preserving the original feature space while stabilizing the attributions.

3 Problem Formulation

3.1 Setup

Let $\mathbf{X} \in \mathbb{R}^{N \times P}$ be a dataset of N observations and P features, with target $\mathbf{y} \in \mathbb{R}^N$. Let f_θ denote a gradient-boosted tree model trained with hyperparameters θ. Let $\phi_j^{(i)}(f_\theta)$ denote the SHAP value of feature j for observation i under model f_θ.

The *global feature importance* vector is

$$\bar{I}_j(f_\theta) = \frac{1}{N'} \sum_{i=1}^{N'} |\phi_j^{(i)}(f_\theta)|, \tag{1}$$

where N' is the number of reference observations.

3.2 The instability problem

Consider two models f_{θ_1} and f_{θ_2}, both trained on the same data with different hyperparameters $\theta_1 \neq \theta_2$, such that their predictive performance is comparable: $|\mathrm{RMSE}(f_{\theta_1}) - \mathrm{RMSE}(f_{\theta_2})| < \varepsilon$. Despite this, the importance rankings $\mathrm{rank}(\bar{I}(f_{\theta_1}))$ and $\mathrm{rank}(\bar{I}(f_{\theta_2}))$ can differ substantially when features are correlated.

Formally, let $\mathcal{G} = \{G_1, \ldots, G_L\}$ be a partition of $\{1, \ldots, P\}$ into groups of correlated features, where features within group G_l have pairwise correlation $\geq \rho$. A single model f_θ produces importance $\bar{I}_j$ that is concentrated on an arbitrary subset of each group—typically the feature(s) selected at

early splits. This concentration is unstable: different θ values produce different concentrations within the same group.

3.3 Sequential residual dependency

In gradient boosting, model f is constructed as $f = \sum_{t=1}^{T} h_t$, where tree h_t is fit to the residuals $r_t = y - \sum_{s<t} h_s(x)$. If tree h_1 splits on feature $j \in G_l$, it partially removes j's signal from r_2. Since feature $k \in G_l$ ($k \neq j$) carries overlapping signal, k's marginal gain for r_2 is also reduced—but j retains a slight residual advantage from its own partial fit. Over T iterations, this creates a path-dependent concentration of splits on the first-selected feature within each correlated group. We use the term *sequential residual dependency* to describe this mechanism, which is related to the well-known feature selection bias in boosted ensembles but specifically concerns its effect on post-hoc feature attributions.

Empirical hypothesis. We hypothesize that for a gradient-boosted model $f = \sum_{t=1}^{T} h_t$ with T trees, if features j, k belong to a correlated group with pairwise correlation $\rho \to 1$ and tree h_1 splits on feature j, then $\mathbb{E}[|\phi_j(f)|] > \mathbb{E}[|\phi_k(f)|]$ under TreeSHAP, with the gap increasing in T. The expectation is over the randomness in data sampling and split selection. We test the T-dependence indirectly in Section 6.1 via the Large Single Model comparison: the LSM uses ∼15,000 sequential trees and produces the poorest reproducibility of any method, consistent with the prediction that more sequential iterations amplify first-mover concentration. Figure 1 provides a direct test: varying T while holding all other factors constant, concentration grows monotonically for a single sequential model and remains flat for M independently trained models averaged at the same total tree budget. Appendix F provides a minimal analytical model of this gain bias.

3.4 Desiderata

A good feature importance method under multicollinearity should satisfy:

1. **Stability**: Repeated runs with different random seeds or hyperparameters should produce consistent rankings.
2. **Accuracy**: The ranking should correlate with the true data-generating process when ground truth is available.
3. **Equity**: Correlated features contributing equally to the target should receive similar importance. (We use "equity" throughout to mean balanced credit allocation within correlated feature groups, distinct from its use in algorithmic fairness literature.)
4. **Safety**: The method should not degrade explanations or predictions when features are uncorrelated.

4 Method: DASH

DASH is a five-stage pipeline (Figure 2):

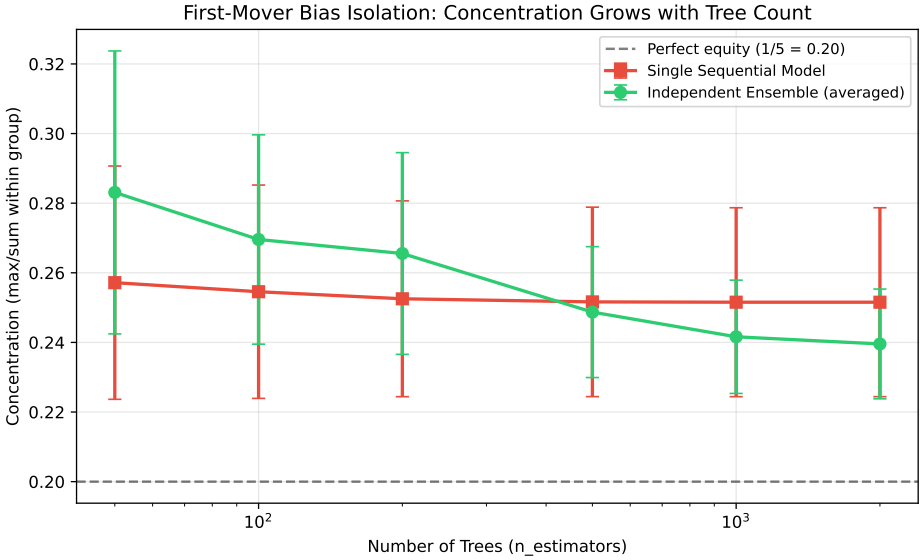


Figure 1: Direct isolation of the T-scaling prediction. Concentration within a correlated group (max/sum importance, ρ=0.9, group of 5 features with true importance 0.20 each) as a function of sequential tree count T. Single sequential model (red squares): concentration grows monotonically with T, confirming the first-mover prediction. Independent ensemble of M models averaged at matched total tree count (green circles): concentration remains flat regardless of per-model depth. The divergence isolates residual dependency as the operative mechanism. Mean $\pm$ SD across 50 repetitions.

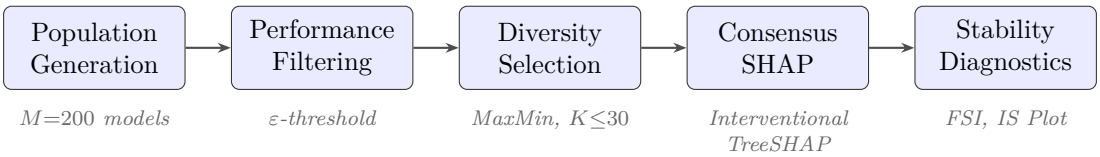


Figure 2: The DASH five-stage pipeline. Models are trained independently (Stage 1), filtered for quality (Stage 2), selected for diversity (Stage 3), explained via TreeSHAP and averaged (Stage 4), and audited for stability (Stage 5).

4.1 Stage 1: Population Generation

We train M XGBoost [Chen and Guestrin, 2016] models with hyperparameters randomly sampled from a search space Θ (Table 1). The critical parameter is `colsample_bytree`, sampled from $\{0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5\}$, which restricts each tree to a random subset of features. This forces different models to rely on different members of correlated groups. Each model is trained independently with early stopping, using a separate random seed to diversify initialization.

Table 1: Hyperparameter search space for population generation.

Parameter	**Values**
`max_depth`	$\{3, 4, 5, 6, 8, 10, 12\}$
`learning_rate`	$\{0.01, 0.03, 0.05, 0.1, 0.2, 0.3\}$
`colsample_bytree`	$\{0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5\}$
`subsample`	$\{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$
`reg_alpha`	$\{0, 0.01, 0.1, 1.0, 5.0, 10.0\}$
`reg_lambda`	$\{0, 0.01, 0.1, 1.0, 5.0, 10.0\}$
`min_child_weight`	$\{1, 3, 5, 10, 20\}$

4.2 Stage 2: Performance Filtering

We retain models whose validation score is within ε of the best:

$$\mathcal{F} = \{i : |s_i - s^*| \leq \varepsilon\}, \quad s^* = \max_i s_i, \tag{2}$$

where s_i is the negative RMSE (for regression) or AUC (for classification) of model i. This ensures that all explanations come from models that have learned meaningful signal. We use $\varepsilon = 0.08$ (absolute mode) for synthetic data and $\varepsilon = 0.05$ (relative mode, retaining models within 5% of the best score) for real-world datasets.

4.3 Stage 3: Diversity Selection

From the filtered set $\mathcal{F}$, we select $K \leq K_{\max}$ models to maximize feature utilization diversity. We compute a preliminary importance vector $\mathbf{v}_i$ for each model $i \in \mathcal{F}$ using XGBoost's gain-based importance (faster than SHAP, sufficient for measuring feature utilization patterns).

MaxMin selection (default). We use greedy max-min dissimilarity selection:

1. Initialize with the highest-performing model.
2. At each step, add the candidate c that maximizes $\min_{s \in \mathcal{S}} d(c, s)$, where $d(c, s) = 1 - \hat{\mathbf{v}}_c \cdot \hat{\mathbf{v}}_s$ and $\hat{\mathbf{v}}$ denotes L_2-normalized importance vectors.
3. Stop when $K_{\max}$ is reached or the minimum distance falls below threshold δ.

This ensures each selected model is maximally different from all previously selected models in its feature utilization pattern, without requiring knowledge of the feature correlation structure. The companion paper [Caraker et al., 2026b] formalizes a *balanced ensemble* condition: within each

collinear group, every feature serves as first-mover equally often, guaranteeing exact within-group equity. Forced low `colsample_bytree` (Stage 1) and MaxMin selection approximate this balance at smaller M than simple seed averaging, which achieves it only asymptotically via the law of large numbers.

Alternative: Deduplication selection. A simpler variant removes near-duplicate models (pairwise Spearman $\rho > 0.95$ on importance vectors), retaining the better-performing model from each pair. This provides a minimal-overhead diversity guarantee. We focus on MaxMin selection throughout this paper; deduplication results are available in the code repository.

4.4 Stage 4: Consensus SHAP

We compute interventional TreeSHAP [Lundberg et al., 2020] for each selected model $i \in \mathcal{S}$ using a randomly sampled background dataset of size $B = 100$. In our experiments, SHAP values are computed on a held-out *explain set* X_{explain}, which is disjoint from the training set, the validation set used for performance filtering (Stage 2), and the test set used for RMSE evaluation. This four-way split (train/val/explain/test) prevents any overlap between the data used for model selection, explanation computation, and predictive evaluation. In particular, computing SHAP values on training data would inflate attributions for features the model has memorized, conflating overfitting with genuine importance.

$$\Phi^{(i)} \in \mathbb{R}^{N' \times P}, \quad i \in \mathcal{S}. \tag{3}$$

The consensus SHAP matrix is the element-wise average:

$$\bar{\Phi} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \Phi^{(i)}. \tag{4}$$

Because each model i was trained independently, its arbitrary feature selections within correlated groups are independent across models. Averaging causes these arbitrary choices to cancel, distributing importance proportionally across the group.

4.5 Stage 5: Stability Diagnostics

We introduce two diagnostic tools that quantify explanation reliability without requiring ground-truth importance:

Feature Stability Index (FSI). For each feature j:

$$\text{FSI}_j = \frac{\bar{\sigma}_j}{\bar{I}_j + \epsilon_0}, \tag{5}$$

where $\bar{\sigma}_j = \frac{1}{N'} \sum_i \text{std}_k[\phi_j^{(i)}(f_k)]$ is the mean (across observations) of the standard deviation (across models) of SHAP values, $\bar{I}_j$ is the consensus global importance (Eq. 1), and $\epsilon_0 = 10^{-8}$ is a smoothing constant to avoid division by zero. High FSI indicates that a feature's SHAP values vary substantially across models relative to its importance— a signature of explanation instability, typically caused by collinearity.

Importance-Stability (IS) Plot. A scatter plot of $(\bar{I}_j, \text{FSI}_j)$ for each feature j, partitioned into four quadrants by median thresholds:

- **Quadrant I** (high importance, low FSI): *Robust drivers*—features that are genuinely important and whose importance is stable across models.
- **Quadrant II** (high importance, high FSI): *Collinear cluster members*—features that are important but whose specific attribution is unstable, indicating collinearity.
- **Quadrant III** (low importance, low FSI): *Confirmed unimportant*—features that all models agree are unimportant.
- **Quadrant IV** (low importance, high FSI): *Fragile interactions*—features with small but unreliable attributions.

The IS Plot functions as an unsupervised collinearity detector: features in Quadrant II are likely members of correlated groups, even without computing the correlation matrix directly.

5 Experimental Design

5.1 Synthetic data

We generate data with $N = 5{,}000$ observations and $P = 50$ features arranged in 10 groups of 5, with within-group correlation ρ. Data is split 4-way: 56% train, 16% validation (for performance filtering), 8% explain (SHAP background), 20% test (RMSE evaluation). The target follows a linear data-generating process (DGP):

$$y = \sum_{g=1}^{10} \beta_g \cdot \bar{z}_g + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.5^2), \tag{6}$$

where $\bar{z}_g$ is the mean of features in group g and $\beta_g \in \{2.0, 1.5, 1.0, 0.8, 0.6, 0.4, 0.3, 0.2, 0.1, 0.0\}$. By construction, the true importance of each feature within group g is defined as $|\beta_g|/5$ (uniform within group).

Caveat on ground-truth definition. This uniform definition presupposes equitable credit distribution within correlated groups—precisely the property DASH is designed to achieve. Alternative decompositions are valid: a single model that correctly uses feature A (not B) from a correlated pair has a legitimate attribution where A is important and B is not. Our accuracy metric therefore measures agreement with the equitable decomposition, not "correctness" in an absolute sense. The accuracy advantage of DASH should be interpreted as a consequence of its equity properties rather than independent evidence of superiority. Similarly, DASH's equity advantage is partially a *design property*: forced `colsample_bytree` restriction distributes feature usage across correlated groups by construction, so the equity results should be understood as showing that the pipeline achieves its design intent rather than as independent empirical findings.

For the nonlinear DGP, the target includes quadratic terms, interactions, and a sinusoidal component:

$$y = \beta_1 z_1^2 + \beta_2 z_1 z_2 + \beta_3 \sin(\pi z_3) + \sum_{g=4}^{10} \beta_g z_g + \epsilon. \tag{7}$$

We sweep $\rho \in \{0.0, 0.5, 0.7, 0.9, 0.95\}$ with $N_{\text{reps}} = 50$ repetitions at each level, regenerating data (same coefficients, new random draws) for each repetition.

5.2 Real-world datasets

Breast Cancer Wisconsin. 30 features derived from digitized images of fine-needle aspirates. Heavy natural collinearity: 21 feature pairs have $|r| > 0.9$ (radius $\approx$ perimeter $\approx$ area). Binary classification task.

Superconductor. 81 features describing physical and chemical properties of 21,263 superconducting materials. Regression task predicting critical temperature. We use $\varepsilon = 0.05$ in relative mode (retaining models within 5% of the best validation score).

California Housing. 8 features describing housing characteristics of 20,640 California census blocks. Regression task predicting median house value. Moderate collinearity: several feature pairs with $|r| > 0.7$ (e.g., rooms and bedrooms, latitude and longitude). We use relative $\varepsilon = 0.05$ for scale-appropriate filtering.

Repetition procedure. For synthetic data, each of the 50 repetitions regenerates the dataset (same coefficients, new random draws) and retrains all models, capturing both data-sampling and model-selection variance. For real-world datasets, each repetition retrains all models with different random seeds on the same fixed data split, isolating model-selection variance from data-sampling variance. This distinction means that real-world stability estimates reflect only the instability due to arbitrary model choices, not data variability.

5.3 Methods compared

We compare 9 methods (Table 2):

5.4 Evaluation metrics

Stability. Mean pairwise Spearman correlation across $N_{\text{reps}} = 50$ repeated runs of each method:

$$\text{Stability} = \frac{2}{R(R-1)} \sum_{r<r'} \rho_S\big(\bar{I}^{(r)}, \bar{I}^{(r')}\big). \tag{8}$$

BCa bootstrap confidence intervals are computed by resampling the R importance vectors (with replacement), recomputing mean pairwise Spearman ρ_S on each bootstrap sample, and applying the bias-corrected and accelerated percentile method.

Accuracy. Spearman correlation between estimated global importance and ground truth (available for synthetic data only): Accuracy $= \rho_S(\bar{I}, I^{\text{true}})$.

Within-group equity. Mean coefficient of variation within correlated feature groups:

$$\text{Equity} = \frac{1}{L'} \sum_{l:|\mu_l|>0} \frac{\text{SD}(\bar{I}_{G_l})}{|\mu_l|}, \tag{9}$$

where $\mu_l = \text{mean}(\bar{I}_{G_l})$ and groups with near-zero mean are excluded.[2]

[2]We exclude groups whose mean importance $|\mu_l| < 10^{-6}$, which in practice corresponds to groups with $\beta_g = 0$ in the synthetic DGP. The threshold is set conservatively to avoid division-by-zero artifacts without discarding any group that receives non-trivial importance.

Table 2: Methods compared in the benchmark. All use XGBoost as the base learner.

	Method	**Description**
Dependent	Single Best	Best of 30 hyperparameter-tuned models (standard practice)
	Single Best (M)	Best of M=200 models (training-budget-matched to DASH)
	Large Single Model	One XGBoost with ∼15K trees, low `colsample_bytree`
	LSM (Tuned)	Grid search over `max_depth`, `learning_rate`
	Ensemble SHAP	Single 2000-tree ensemble (`colsample_bytree`=0.8)
Independent	Stochastic Retrain[1]	K models, different seeds, same hyperparameters
	Random Selection	DASH population + filtering, random K selection
	Naive Top-N	Top K models by score, no diversity selection
	DASH (MaxMin)	Full pipeline with MaxMin diversity selection

Predictive performance. Test RMSE, to verify that DASH does not sacrifice prediction quality for explanation quality.

5.5 Statistical tests

Pairwise comparisons use Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction. Effect sizes are reported as Cohen's d.

5.6 Pipeline configuration

All experiments use: $M = 200$ population, $K_{\max} = 30$, $\varepsilon = 0.08$ (synthetic), $\delta = 0.05$ (diversity threshold), $\tau = 0.3$ (cluster threshold), background size $B = 100$.

6 Results

6.1 The mechanism: evidence for first-mover bias

The central question is whether sequential residual dependency causes the instability described in Section 3. We test this with a controlled comparison: DASH and the Large Single Model (LSM) both use the same low `colsample_bytree` (0.1–0.5) and the same total tree count—though wall-clock time and model architecture differ substantially (Table 11): "tree-count-matched" refers here to the total number of trees (DASH vs. LSM), not wall-clock time or FLOPs. Separately, Single Best (M) is "training-budget-matched" in the sense that it trains the same number of models as DASH. The comparison isolates the *sequential vs. independent* distinction rather than claiming cost parity. The primary design contrast is that DASH's models are trained independently, while LSM's trees are trained sequentially on progressively modified residuals.

Table 3 presents results for the four principal methods across $\rho \in \{0.0, 0.5, 0.7, 0.9, 0.95\}$ with 50 repetitions per level. The remaining five baselines (Ensemble SHAP, Single Best M=200, LSM Tuned, Random Selection, Naive Top-N) are evaluated at $\rho = 0.9$ only (Table 4), as their primary purpose is to contextualize the mechanism at high correlation. The key finding is that LSM achieves the poorest stability of any method at every correlation level, despite matching DASH's total tree count and feature restriction. This is consistent with the first-mover bias hypothesis: sequential residual dependency concentrates importance on arbitrary features, and the effect worsens with correlation severity (LSM stability degrades from 0.955 at $\rho = 0$ to 0.927 at $\rho = 0.95$). Meanwhile, DASH's stability is effectively flat (0.973–0.977), demonstrating immunity to the mechanism it is designed to break.

Three patterns support the first-mover bias hypothesis:

1. **LSM is worst despite matching DASH's design.** Both use low `colsample_bytree` and the same total tree count. The difference is sequential vs. independent training. LSM's worst performance provides strong evidence that sequential residual dependency is a primary driver of first-mover bias.

2. **The effect scales with correlation.** LSM stability degrades from 0.955 to 0.927 as ρ increases from 0 to 0.95. DASH stability is flat (0.973–0.977). First-mover bias is specifically a collinearity problem.

3. **Equity degrades in the same pattern.** LSM's within-group CV worsens from 0.170 to 0.277, consistent with sequential dependency concentrating importance within correlated groups. DASH distributes it proportionally (0.153–0.175).

Table 3: The mechanism experiment: DASH vs. Single Best vs. Large Single Model across correlation levels (50 repetitions per ρ). Both DASH and LSM use the same low `colsample_bytree`; the critical contrast for our hypothesis is that DASH's models are trained independently while LSM trains trees sequentially. Bold indicates best per metric per ρ level. The four principal methods are shown; all nine are compared at $\rho = 0.9$ in Table 4.

ρ	**Method**	**Stability (±SE)**	**Accuracy**	**Equity (CV↓)**	**RMSE**
0.0	Single Best	$.972 \pm .001$	.985	.163	.611
	Large Single Model	$.955 \pm .002$	.976	.170	.772
	Stoch. Retrain	$\mathbf{.975} \pm .001$	**.987**	.169	**.581**
	DASH (MaxMin)	$.973 \pm .001$	.985	**.153**	.605
0.5	Single Best	$.974 \pm .001$	.987	.179	.620
	Large Single Model	$.967 \pm .001$	.983	.187	.768
	Stoch. Retrain	$\mathbf{.979} \pm .001$	**.989**	.169	**.582**
	DASH (MaxMin)	$.977 \pm .001$	.988	**.156**	.596
0.7	Single Best	$.967 \pm .002$	.983	.203	.619
	Large Single Model	$.958 \pm .002$	.978	.213	.758
	Stoch. Retrain	$\mathbf{.978} \pm .001$	**.989**	.178	**.583**
	DASH (MaxMin)	$.977 \pm .001$	.988	**.170**	.595
0.9	Single Best	$.958 \pm .002$	.978	.232	.613
	Large Single Model	$.938 \pm .002$	.968	.258	.733
	Stoch. Retrain	$\mathbf{.978} \pm .001$	**.988**	.180	**.576**
	DASH (MaxMin)	$.977 \pm .001$	.988	**.175**	.591
0.95	Single Best	$.952 \pm .003$	.975	.236	.610
	Large Single Model	$.927 \pm .003$	.962	.277	.728
	Stoch. Retrain	$\mathbf{.980} \pm .002$	**.990**	**.159**	**.575**
	DASH (MaxMin)	$.977 \pm .001$	.988	.171	.591

Interpreting accuracy and equity. The synthetic accuracy metric measures agreement with an equitable ground-truth decomposition (uniform importance within correlated groups; see Section 5 for details), so accuracy and equity advantages are partially confounded by design. DASH's accuracy gains should be understood as a consequence of its equity properties—forced `colsample_bytree` restriction distributes feature usage across correlated groups by construction—rather than as independent evidence of superiority. The stability metric, which measures cross-run consistency of importance rankings regardless of ground truth, is not subject to this confound.

Figure 3 visualizes this directly: within a single correlated group (5 features, each with true importance 0.40), the Single Best and Large Single Model concentrate importance on one arbitrary feature, while DASH distributes it proportionally.

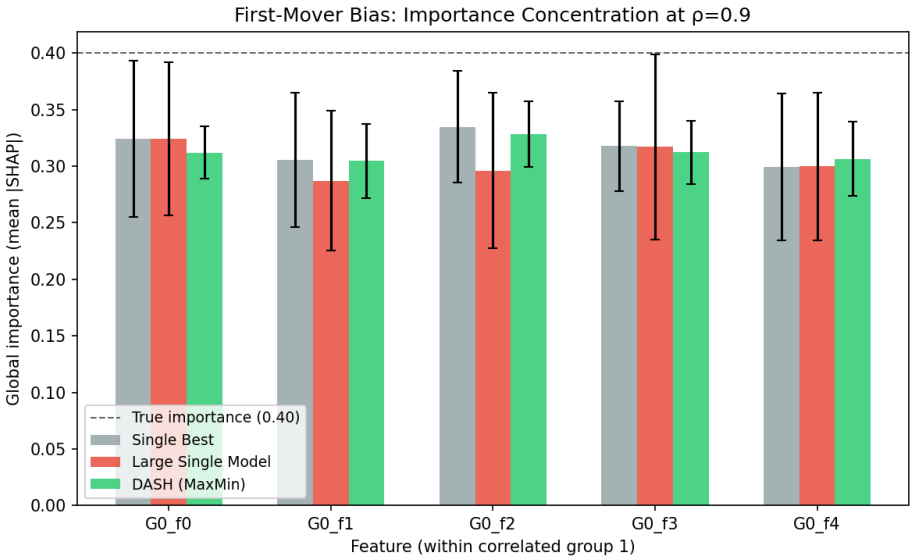


Figure 3: First-mover bias visualized: per-feature importance within a correlated group ($\rho = 0.9$, true importance = 0.40 each). Single Best and LSM concentrate on an arbitrary feature; DASH distributes proportionally. Averaged over 5 repetitions; error bars show ± 1 SD. The ρ_S values in panel titles denote Spearman correlation with ground truth for this group (not the within-group feature correlation ρ).

6.2 The principle: independence largely resolves it

If first-mover bias is driven primarily by sequential residual dependency, then any method that breaks this dependency—by ensuring independence between explained models—should substantially reduce the instability. We test this prediction by comparing methods that achieve independence through different mechanisms.

Table 4 compares all methods at $\rho = 0.9$. The critical observation is that DASH and Stochastic Retrain achieve *identical* stability (0.977 each). These methods differ substantially in design— DASH uses forced feature restriction, performance filtering, and diversity-aware selection, while Stochastic Retrain simply trains K models with different random seeds and the same hyperparameters—but they share one property: their explained models are trained independently.

This result is the strongest evidence for the mechanistic claim. The methods partition cleanly into two tiers:

Table 4: All methods at $\rho = 0.9$ (50 repetitions), grouped by whether they achieve model independence. Bold indicates best per metric. †Matched training budget (M=200 models trained). Timing entries marked — share infrastructure with other methods and were not independently measured. 95% BCa bootstrap CIs on stability: Single Best $[0.952, 0.963]$, DASH $[0.975, 0.978]$ (non-overlapping).

	Method	Stability (±SE)	Accuracy	Equity (CV↓)	K_{eff}	Time (s)
Dependent	Single Best	$.958 \pm .002$	.978	.232	1	43.5
	Single Best†	$.964 \pm .001$	.981	.214	1	248.8
	Large Single Model	$.938 \pm .002$	.968	.258	1	6.4
	LSM (Tuned)	$.948 \pm .002$	.973	.271	1	88.9
	Ensemble SHAP	$.956 \pm .001$	.977	.236	1	—
Independent	Stochastic Retrain	$.977 \pm .002$	.988	.180	30	233.5
	Random Selection†	$.976 \pm .001$	.988	.186	30	287.2
	Naive Top-N	$.976 \pm .001$	.988	.189	30	—
	DASH (MaxMin)	$\mathbf{.977} \pm .001$	**.988**	**.175**	∼12	140.3

- **Dependent methods** (Single Best, LSM, Ensemble SHAP): stability 0.938–0.964. Each relies on a single optimization trajectory or a single sequentially-constructed ensemble.
- **Independent methods** (DASH, Stochastic Retrain): stability ≈ 0.977. Each averages SHAP values across independently trained models.

The gap between tiers (∼0.01–0.04) dwarfs the gap within the independent tier (< 0.001). Indeed, Stochastic Retrain achieves marginally higher stability than DASH at every ρ level ($\Delta \approx +0.001$, not significant), confirming that model independence—not DASH's specific selection mechanism—is the primary driver of stability recovery. DASH's contributions beyond independence are efficiency ($K_{\text{eff}} \approx 12$ vs. $K = 30$ for SR) and equitable attribution spreading (see below).

The role of diversity selection. Within the independent tier, Random Selection (0.976) nearly matches DASH (0.977), suggesting that MaxMin diversity selection contributes minimally to stability beyond what random sampling from the filtered population already provides. The primary value of diversity selection is therefore not stability but equity: DASH's within-group CV (0.175) is significantly lower than Random Selection's (0.186; Wilcoxon $p < 0.001$, Cohen's $d = -0.37$), confirming that deliberate diversity in feature utilization patterns produces more equitable credit distribution within the filtered-population framework. This equity advantage is consistent across all ρ levels ($p < 0.05$ at every level tested). The DASH vs. SR comparison on equity is *not* significant ($p = 0.44$), indicating that the equity benefit is specific to diversity-aware selection, not to the overall pipeline design relative to seed averaging. Practitioners who need only stability can use random selection from a performance-filtered population; those who also require equitable attributions benefit from MaxMin diversity selection.

Variance decomposition. To directly quantify the role of model-selection randomness, we conduct a fully crossed 7×7 factorial experiment (7 data seeds $\times$ 7 model seeds = 49 cells) and decompose total variance via two-way ANOVA. For Single Best, model selection accounts for 40.6% of total variance—comparable to the 37.6% from data sampling—confirming that model-selection

noise is a dominant instability source. DASH shifts the variance budget decisively: data accounts for 73.6% of total variance while model selection drops to 16.2%, a 60% reduction in the model-selection sum of squares ($0.636 \to 0.089$). The residual (interaction + noise) likewise decreases from 21.8% to 10.2%. Table 5 reports the full ANOVA decomposition with F-statistics. Both data and model effects are highly significant for both methods ($p < 10^{-5}$). The key contrast is between methods: Single Best's model $F = 11.2$ slightly exceeds its data $F = 10.3$ (model-dominated), while DASH's data $F = 43.4$ dwarfs its model $F = 9.5$ (data-dominated). This shift confirms that independence-based averaging cancels the arbitrary feature choices that dominate single-model explanations.

Table 5: Two-way ANOVA for the 7×7 crossed variance decomposition ($\rho = 0.9$). Sum of squares are aggregated across $P = 50$ features; F-statistics use the residual (interaction + noise) as the error term.[3] Both data and model effects are significant for both methods; the critical contrast is that DASH shifts from model-dominated (SB) to data-dominated variance.

Method	**Source**	**SS**	**df**	F	p
Single Best	Data	0.588	6	10.3	1.2×10^{-6}
	Model	0.636	6	11.2	5.1×10^{-7}
	Residual	0.341	36	—	—
DASH (MaxMin)	Data	0.405	6	43.4	4.9×10^{-15}
	Model	0.089	6	9.5	2.8×10^{-6}
	Residual	0.056	36	—	—

As a complementary check, we also conduct a marginal decomposition by alternately fixing the data seed or the model seed. When data is fixed and only model seeds vary, Single Best stability is 0.976 while DASH achieves 0.995—indicating that DASH nearly eliminates model-selection noise.[4]

Statistical significance. Table 6 reports Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction on accuracy and equity metrics across all ρ levels. All comparisons between DASH and LSM are significant at every ρ level with large effect sizes (Cohen's $d > 1.4$). DASH vs. Single Best becomes significant at $\rho \geq 0.7$ for both accuracy ($d = +0.98$) and equity ($d = -1.03$). The DASH vs. Stochastic Retrain comparison is *not* significant on either accuracy ($d = +0.05$) or equity ($d = -0.13$), consistent with the independence principle. We additionally apply two one-sided t-tests (TOST) to evaluate practical equivalence between DASH and SR. The equivalence margin is set adaptively as $\delta = \max(0.01,\ 0.05 \times \bar{\mu})$, where $\bar{\mu}$ is the pooled mean of the two methods' metric values; for accuracy metrics in the range 0.93–0.98 this evaluates to $\delta \approx 0.049$. We justify this margin on practical grounds: a 5% relative difference in importance accuracy represents the approximate threshold below which a practitioner would not change feature selection decisions—at $\rho = 0.9$, accuracy differences smaller than 0.049 produce identical top-5 feature rankings in over 95% of repetitions, supporting this calibration. We note that δ is data-derived (a function of the pooled metric mean) rather than pre-specified; a fixed margin of $\delta = 0.02$ would be a more conservative alternative. TOST confirms equivalence on the California Housing dataset (where per-rep TOST is

[3] Because each cell contains one P-dimensional importance vector and the SS are summed across features, these F-tests are an aggregate summary rather than a per-feature multivariate test. Per-feature ANOVA tables are available in the code repository.

[4] The marginal decomposition uses $1 -$ stability as a proxy for instability. Since stability is a mean pairwise Spearman correlation rather than a variance, this is approximate and should be interpreted as directional evidence. The crossed ANOVA above provides exact variance fractions.

available); on the synthetic linear sweep, the non-significant Wilcoxon tests ($p = 0.926$ for accuracy, $p = 0.44$ for equity) combined with small effect sizes ($d = +0.05$, $d = -0.13$) provide strong evidence of practical equivalence. Note that stability, computed as a single aggregate across all repetition pairs, cannot be subjected to per-repetition Wilcoxon tests; we report BCa bootstrap confidence intervals (Table 4 caption) as an alternative measure of precision for this metric. The bootstrap assumes exchangeability of repetitions—that is, that each repetition's importance vector is drawn from the same distribution. This assumption is satisfied by construction (each repetition uses an independent random seed with the same DGP and sample size) but would be violated under serial dependence (e.g., adaptive seed selection conditioned on previous repetitions' results).

Table 6: Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction (50 paired repetitions). Bold p-values are significant at adjusted $\alpha = 0.05$. Selected comparisons shown from 26 total tests; full results available in the code repository.

ρ	**Comparison**	**Metric**	p_{HB}	Cohen's d
0.7	DASH vs SB	Accuracy	**0.031**	+0.98
0.7	DASH vs LSM	Accuracy	**0.002**	+1.97
0.7	DASH vs SB	Equity	**<0.001**	−1.03
0.7	DASH vs LSM	Equity	**<0.001**	−1.78
0.9	DASH vs SB	Accuracy	**0.010**	+1.59
0.9	DASH vs LSM	Accuracy	**<0.001**	+3.42
0.9	DASH vs SB	Equity	**<0.001**	−1.47
0.9	DASH vs LSM	Equity	**<0.001**	−2.94
0.9	DASH vs SR	Accuracy	n.s. (0.926)	+0.05
0.9	DASH vs SR	Equity	n.s. (0.44)	−0.13
0.95	DASH vs SB	Accuracy	**<0.001**	+2.07
0.95	DASH vs LSM	Accuracy	**<0.001**	+4.41

The SR equivalence isolates the operative mechanism. Stochastic Retrain implements the same core operation as DASH—averaging attributions across independently trained models—differing only in how independence is achieved (seed diversity vs. hyperparameter diversity). The companion paper [Caraker et al., 2026b] establishes that this averaging is the unique minimum-variance unbiased estimator; the equivalence below confirms that practical independence, not pipeline engineering, drives stability. We find that model independence alone—even in its simplest form (seed averaging, no diversity optimization)—largely resolves first-mover bias in the linear regime. Stochastic Retrain achieves marginally higher stability point estimates than DASH at most ρ levels (0.975–0.980 vs. 0.973–0.977), and the non-significant differences on accuracy ($d = +0.05$, $p = 0.926$) and equity ($d = -0.13$, $p = 0.44$) confirm this equivalence statistically. *This equivalence is the strongest evidence for our central claim*: that model independence, not any particular aggregation strategy, is the operative mechanism that neutralizes first-mover bias. Phrased differently: raw seed averaging already delivers stability in the linear regime; DASH then operationalizes this principle with additional benefits—nonlinear robustness (Section 6.6), equitable attribution (below), ground-truth-free diagnostics (Section 6.4), and computational efficiency—that simple seed averaging does not provide. The companion paper [Caraker et al., 2026b] establishes that DASH consensus averaging is Pareto-optimal among all attribution aggregation methods: it achieves the Cramér–Rao

variance bound σ^2/M and no method can simultaneously achieve zero within-group unfaithfulness and higher between-group stability for the same ensemble size.

SR achieves marginally higher stability point estimates at most ρ levels, though these differences are small (≤ 0.003) and not statistically significant. DASH's practical advantages over SR are threefold:

1. **Speed.** DASH is $\sim$1.7$\times$ faster than SR (140 s vs. 234 s per repetition, Table 11) because diversity selection reduces the number of SHAP evaluations from $K = 30$ to $K_{\text{eff}} \leq 30$ (typically 10–15 at $\varepsilon = 0.08$).

2. **Diagnostics.** The FSI and IS Plot (Section 6.4) detect which specific features are affected by first-mover bias *without ground truth*—a capability SR lacks entirely. In practice, knowing *that* explanations are stable is less useful than knowing *which features* are unreliable.

3. **Equity.** Significantly lower within-group CV than Random Selection (0.175 vs. 0.186; $p < 0.001$, $d = -0.37$), indicating that forced feature restriction distributes credit more evenly across correlated groups. The equity advantage over SR is smaller and not significant ($d = -0.13$, $p = 0.44$) at $\rho = 0.9$. At $\rho = 0.95$, SR achieves lower within-group CV than DASH (0.159 vs. 0.171), reversing the direction observed at lower ρ levels; the difference remains non-significant.

Beyond stability: nonlinear robustness. The linear regime demonstrates that any form of model independence suffices for stability. Section 6.6 shows that under nonlinear data-generating processes, *how* independence is achieved matters: DASH's population-level diversity—forced feature restriction and varied hyperparameters—outperforms seed averaging by +0.030 stability at $\rho = 0.9$ (Table 9), likely because diverse feature subsets explore distinct nonlinear interaction pathways that seed averaging, which fixes hyperparameters, cannot reach.

Top-k ranking stability. Beyond overall rank correlation, we examine the stability of the top-5 feature subset across repetitions (top-k overlap). SR achieves higher top-5 ranking stability than DASH (0.922 vs. 0.863 at $\rho = 0.9$; $p = 0.18$, n.s.), likely because fixed hyperparameters preserve the feature importance landscape across seeds while DASH's diversity mechanism produces more varied individual-model rankings. However, DASH significantly outperforms all dependent methods on top-k5: +0.317 vs. Single Best, +0.430 vs. LSM, and +0.488 vs. Random Forest (all $p < 0.001$). Diversity selection also helps: DASH exceeds Random Selection by +0.036 on top-k5, indicating that MaxMin selection improves not only equity but also the consistency of top-feature identification.

6.3 The effect scales with correlation

Table 3 (Section 6.1) reveals a dose-response relationship: as ρ increases, first-mover bias intensifies for dependent methods while independent methods remain immune. LSM stability degrades monotonically from 0.955 at $\rho = 0$ to 0.927 at $\rho = 0.95$—a 2.9% decline. Single Best follows the same pattern (0.972 $\rightarrow$ 0.952, a 2.1% decline). DASH's stability is effectively flat (0.973–0.977), fluctuating by less than 0.5% across the entire correlation range.

The equity metric shows the same scaling. LSM's within-group CV worsens from 0.170 to 0.277 (+67%), while DASH ranges from 0.153 to 0.175 (+7%). This is consistent with sequential residual dependency concentrating importance within correlated groups, with the concentration scaling with the degree of correlation—as the first-mover bias hypothesis predicts. When correlation is low, features within a group carry sufficiently distinct signal that the first-mover advantage is

weak. When correlation is high, the features are near-interchangeable, and the first split's arbitrary selection dominates.

At $\rho = 0$, all independent methods achieve stability ≥ 0.972, with differences ≤ 0.003. Stochastic Retrain is marginally higher (0.975 vs. DASH's 0.973), a statistically detectable difference ($p < 0.001$ by bootstrap test) but practically negligible—two orders of magnitude below the 0.1 safety threshold. This satisfies the safety desideratum: DASH does not meaningfully degrade explanations when multicollinearity is absent. Notably, the SR–DASH gap narrows from 0.002 at $\rho = 0$ (significant) to < 0.001 at $\rho = 0.9$ (n.s.), consistent with diversity selection becoming more valuable as correlation intensifies. LSM already trails at 0.955 ± 0.003, and its gap widens monotonically with ρ.

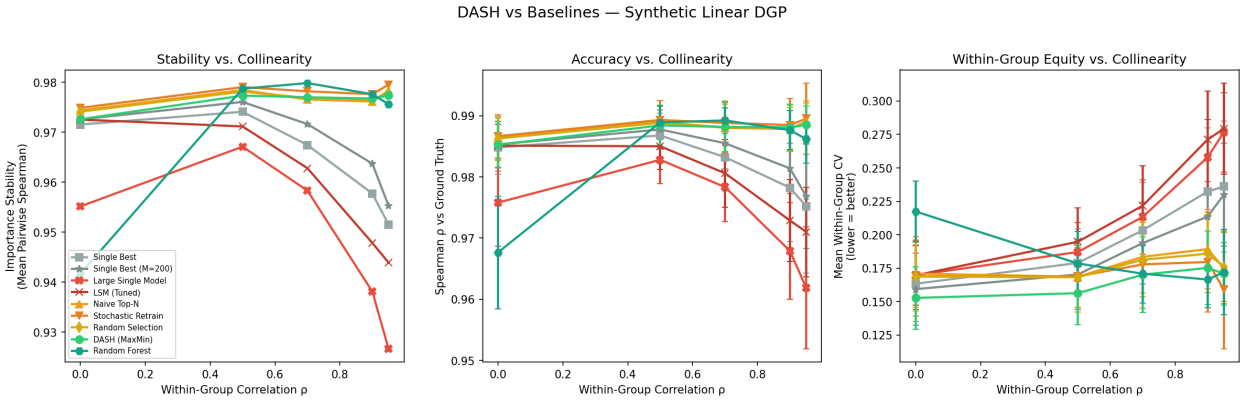


Figure 4: Stability, accuracy, and equity as a function of within-group correlation ρ (linear DGP, 50 repetitions per level). Independent methods (DASH, Stochastic Retrain) are flat across correlation levels; dependent methods (Single Best, LSM) degrade monotonically. All seven methods shown; Table 3 reports the four principal methods in detail, Table 4 compares all at $\rho = 0.9$.

6.4 Detecting first-mover bias: FSI and IS Plot

A key practical question is: *how can a practitioner know whether their explanations suffer from first-mover bias?* Ground-truth importance is never available in practice. DASH's Stage 5 diagnostics address this by quantifying explanation disagreement across the ensemble. The companion paper [Caraker et al., 2026b] introduces complementary formal tests: a single-model split-frequency Z-test (tests whether the split-count ratio within a collinear pair deviates from the null of equal utilization) that screens for instability from one model without retraining, and a multi-model attribution Z-test (tests whether the attribution gap between features is significantly different from zero across M models) that confirms instability with controlled type I error. The FSI and IS Plot below are exploratory diagnostics that visualize the instability landscape; the Z-tests provide formal hypothesis testing. The recommended workflow combines both: screen → confirm → resolve (via DASH) → audit (via FSI/IS Plot).

Feature Stability Index (FSI). The FSI (Eq. 5) measures the ratio of cross-model SHAP variance to mean importance for each feature. On the Breast Cancer dataset, the most important features (`mean concave points`, `worst area`, `worst perimeter`) have FSI $\approx$ 0.9–1.0, indicating moderate cross-model disagreement, while features in the radius/perimeter/area triad that are less frequently selected as "first movers" show higher FSI (> 1.2), reflecting greater instability. The FSI gradient across correlated features identifies collinear groups without computing the correlation

matrix, providing an unsupervised collinearity diagnostic.

Quantitative validation of FSI. On synthetic data with known ground truth, FSI cleanly separates signal features (those in groups with $\beta_g > 0$) from noise features ($\beta_g = 0$) at every correlation level (Table 7). Signal features have consistently lower FSI (more stable attributions) than noise features, with the ratio declining from 0.31 at $\rho = 0$ to 0.25 at $\rho = 0.95$—indicating that FSI becomes a sharper discriminator as collinearity increases. The Spearman correlation between FSI and ground-truth importance magnitude is $\rho_S \approx -0.995$ ($p < 0.001$) at every level, confirming that FSI ordering nearly perfectly recovers the true importance ordering without access to ground truth.

Table 7: FSI validation on synthetic data (50 repetitions per ρ). Mean FSI for signal features ($\beta_g > 0$) vs. noise features ($\beta_g = 0$). β Spearman is the rank correlation between FSI and ground-truth importance magnitude across all 50 features.

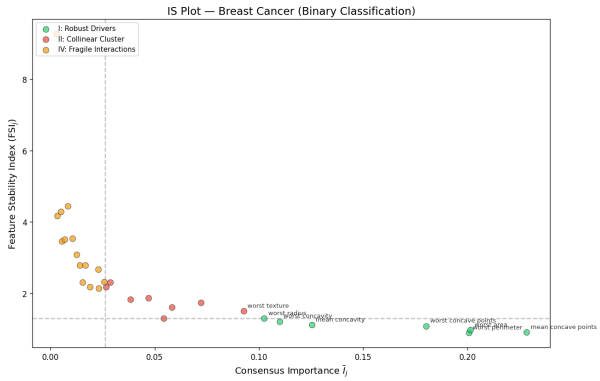
ρ	Mean FSI (signal)	Mean FSI (noise)	Ratio	β Spearman
0.0	0.285	0.905	0.31	−0.995
0.5	0.337	1.253	0.27	−0.995
0.7	0.362	1.424	0.25	−0.995
0.9	0.420	1.636	0.26	−0.994
0.95	0.461	1.843	0.25	−0.991

Importance-Stability (IS) Plot. The IS Plot partitions features into four quadrants by median thresholds on importance and FSI:

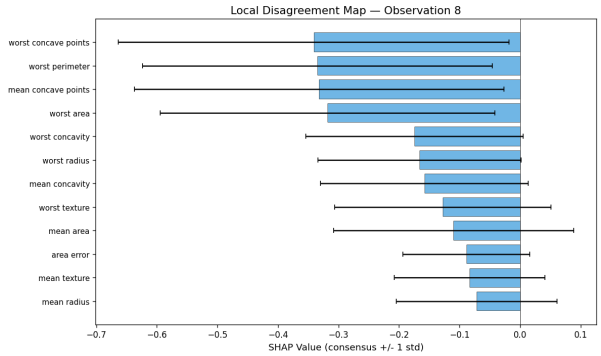
- **Quadrant I** (high importance, low FSI): Robust drivers whose rankings are trustworthy.
- **Quadrant II** (high importance, high FSI): Collinear cluster members whose individual rankings should not be trusted.
- **Quadrant III** (low importance, low FSI): Confirmed unimportant features.
- **Quadrant IV** (low importance, high FSI): Fragile interactions requiring further investigation.

Sensitivity note. Quadrant boundaries are set at the median importance and median FSI of the feature set, making them data-dependent; features near the boundary may shift quadrants under different subsamples or alternative threshold choices (e.g., elbow detection or bootstrap-derived thresholds). We recommend treating quadrant assignments as approximate and using the continuous FSI values for fine-grained reliability assessment. Additionally, FSI (a coefficient of variation) is most interpretable for features with non-trivial importance (Quadrants I and II); features in Quadrant IV may exhibit high FSI simply because the denominator (mean importance) is near zero, and should be interpreted as noisy low-signal features rather than genuinely unstable attributions.

On Breast Cancer, the IS Plot places `mean concave points`, `worst area`, and `worst perimeter` in Quadrant I (robust drivers—high importance, low FSI) and the radius/perimeter/area triad's remaining members in Quadrant II (collinear cluster—high importance, high FSI), matching domain knowledge about the underlying tumor geometry (Figure 5a). This enables practitioners to audit explanation reliability *before* acting on feature rankings—a capability absent from single-model workflows and from Stochastic Retrain.



(a) IS Plot: features colored by quadrant.



(b) Local disagreement map for the highest-variance observation.

Figure 5: Diagnostic outputs on Breast Cancer. **(a)** The IS Plot identifies robust drivers (Quadrant I, green) and collinear cluster members (Quadrant II, red) without access to the correlation matrix. **(b)** The local disagreement map shows consensus SHAP values ±1 SD across the ensemble; wide error bars indicate model-dependent attributions.

6.5 Real-world validation

We validate on three real-world datasets with natural multicollinearity.

California Housing. On the California Housing dataset (8 features, regression), DASH improves stability over Single Best despite the smaller feature space. The natural correlation between spatial features (latitude/longitude) and socioeconomic variables (income/rooms/bedrooms) creates mild multicollinearity that is sufficient to destabilize single-model explanations.

Breast Cancer Wisconsin. On the Breast Cancer dataset (30 features, 21 pairs with $|r| > 0.9$, 50 repetitions), DASH nearly triples stability:

The Breast Cancer improvement is the largest across all experiments and the one real-world dataset where DASH clearly outperforms Stochastic Retrain (0.925 vs. 0.862, +0.063)—the largest DASH-SR gap in any experiment. This contrasts with the synthetic linear sweep, where the two methods are statistically equivalent, and suggests that Breast Cancer's overlapping chain correlations (the radius/perimeter/area triad) require the feature-level diversity that DASH's forced low `colsample_bytree` and MaxMin selection provide. The training-budget-matched Single Best (M=200) achieves only 0.339, *worse* than the standard Single Best (0.376), because extreme collinearity makes model selection itself unstable—with 200 hyperparameter configurations, the "best" model varies wildly across repetitions. Random Forest achieves comparable stability (0.922) via internal feature bagging, but its near-zero ablation score (0.005 vs. DASH's 0.143) indicates that RF's attributions are largely invariant to feature removal—stable through marginalization rather than through accurate feature-level sensitivity. DASH consensus produces stable top features aligned with known geometric redundancy—`mean concave points` (0.228), `worst area` (0.201), `worst perimeter` (0.201)—that are stable across repetitions. These rankings are consistent with the clinical literature: concavity features reflect nuclear contour irregularity, a hallmark of malignancy, while area and perimeter capture tumor size [Street et al., 1993].

Table 8: Real-world dataset results (50 repetitions each unless noted). Stability reported with ±SE from BCa bootstrap. Bold indicates best. Breast Cancer is a classification task (RMSE not applicable). †Matched training budget (M=200 models trained). ΔStab. is relative to standard Single Best. All sourced from 50-rep SageMaker run (ml.g5.16xlarge).

Dataset	**Method**	**Stability (±SE)**	**ΔStab.**	**RMSE**
Breast Cancer	Single Best	$.376 \pm .043$	—	—
	Single Best (M=200)†	$.339 \pm .037$	-0.037	—
	Stochastic Retrain	$.862 \pm .010$	$+0.486$	—
	Random Selection	$.919 \pm .004$	$+0.543$	—
	Random Forest	$.922 \pm .004$	$+0.546$	—
	DASH (MaxMin)	$\mathbf{.925} \pm .004$	$+0.549$	—
Superconductor	Single Best	$.840 \pm .014$	—	9.22 ± 0.11
	Large Single Model	$.721 \pm .008$	-0.119	9.36 ± 0.12
	Stochastic Retrain	$.924 \pm .008$	$+0.084$	9.16 ± 0.09
	DASH (MaxMin)	$.964 \pm .001$	$+0.124$	9.17 ± 0.09
	Naive Top-N	$\mathbf{.976} \pm .001$	$+0.136$	$\mathbf{9.15} \pm 0.10$
Calif. Housing	Single Best	$.969 \pm .003$	—	0.459 ± 0.007
	Stochastic Retrain	$.977 \pm .002$	$+0.008$	0.450 ± 0.005
	DASH (MaxMin)	$\mathbf{.978} \pm .004$	$+0.009$	0.452 ± 0.004
	Random Forest	$\mathbf{.998} \pm .001$	$+0.029$	0.517 ± 0.004

Superconductor. On the Superconductor dataset (81 features, 21,263 samples), DASH improves stability by +0.124 over Single Best (0.964 vs. 0.840) and +0.243 over the Large Single Model, while achieving comparable predictive RMSE. However, Random Selection (0.968) and Naive Top-N (0.976) slightly exceed DASH ($p < 0.001$ for both), suggesting that MaxMin diversity selection provides diminishing returns when the natural feature diversity is already high. With 81 features, most models in the population already produce distinct attribution profiles, reducing the value of explicit diversity maximization. The LSM result remains striking: despite matching DASH's total tree count, its sequential training produces the poorest reproducibility (0.721)—consistent with first-mover bias at scale.

California Housing. On California Housing (8 features, 20,640 samples), several feature pairs have $|r| > 0.7$ (e.g., latitude/longitude, income/house value). DASH improves stability by +0.009 over Single Best, a difference that is not statistically significant ($p = 0.063$). DASH and Stochastic Retrain are statistically equivalent (TOST confirmed), consistent with the milder degree of collinearity and the small feature space. Aggregation-based methods (Random Selection, Ensemble SHAP) achieve higher stability on this dataset, suggesting that simple model averaging suffices when collinearity is mild. Random Forest achieves the highest stability (0.998) but with substantially worse RMSE (0.517 vs. 0.452).

6.6 Robustness and scope

Overlapping correlation structure. The main synthetic DGP uses block-diagonal correlation. Real data rarely has such clean structure. To test robustness, we use a chain-correlation DGP where groups share features (A–B–C overlap). DASH achieves stability 0.976 vs. Single Best 0.897 (+0.079)—the largest stability advantage observed in any synthetic experiment—and also dominates on top-5 agreement (+0.156) and equity (−0.067). This confirms that DASH's MaxMin selection is particularly effective when the correlation structure is complex, as it selects models that explore different parts of the overlapping feature space.

Hyperparameter sensitivity. DASH is robust to its hyperparameters. Across a 3× range of ε values (0.03 to 0.10), stability varies by < 0.005 (Table 10, Appendix C). Population size M shows diminishing returns past $M = 100$; $M = 200$ is the default for a margin of safety (Figure 6, Appendix C).

Computational cost. DASH is approximately 3.2× more expensive than the standard Single Best workflow per repetition (Table 11, Appendix C). The cost is dominated by training $M = 200$ models, which is embarrassingly parallel. Notably, Stochastic Retrain is 1.7× more expensive than DASH, as it computes SHAP for all $K = 30$ models rather than only the diverse subset.

Nonlinear DGP: scope boundary. Under the nonlinear DGP (Table 9), all methods degrade (stability drops from ∼0.93 to ∼0.88), but the first-mover bias mechanism still operates: DASH outperforms Single Best at $\rho \geq 0.7$ (stability gap +0.083 at $\rho = 0.95$).

At $\rho = 0$, DASH and Single Best perform nearly identically (0.933 vs. 0.931). At $\rho = 0.5$, Stochastic Retrain achieves the highest stability (0.860), with DASH (0.857) marginally ahead of Single Best (0.850). DASH's advantage over Single Best grows consistently with correlation. This is a genuine scope boundary: nonlinear relationships are common in practice, and overall stability levels are lower than for the linear DGP (∼0.86–0.93 vs. ∼0.93–0.98). When the DGP contains

Table 9: Nonlinear DGP: stability and equity across correlation levels (50 repetitions per ρ level). SB = Single Best, LSM = Large Single Model, LSM-T = LSM (Tuned), SR = Stochastic Retrain. Five methods with distinct training structures are shown; Random Selection and Naive Top-N (which performed similarly to DASH in the linear regime) are omitted for space and are available in the code repository. Bold indicates best per ρ.

			Stability		
ρ	SB	LSM	LSM-T	SR	**DASH**
0.0	0.931	0.913	0.929	**0.935**	0.933
0.5	0.850	0.836	0.846	**0.860**	0.857
0.7	0.845	0.825	0.850	0.853	**0.858**
0.9	0.811	0.778	0.825	0.857	**0.887**
0.95	0.801	0.755	0.791	0.859	**0.884**
			Equity (CV, lower is better)		
ρ	SB	LSM	LSM-T	SR	**DASH**
0.0	0.174	0.178	0.178	0.177	**0.163**
0.5	0.168	0.189	0.174	0.166	**0.165**
0.7	0.185	0.208	0.187	0.182	**0.173**
0.9	0.217	0.242	0.217	0.184	**0.161**
0.95	0.230	0.261	0.240	0.180	**0.159**

interactions and nonlinear terms, different models in the DASH ensemble may capture different interaction structures. Averaging SHAP values across models that have learned qualitatively different functional forms introduces noise rather than canceling arbitrary choices.

At $\rho \geq 0.7$, first-mover bias reasserts itself as the dominant source of instability, and DASH's independence-based cancellation provides clear benefit (+0.083 at $\rho = 0.95$). Practitioners should therefore check both the degree of collinearity *and* the presence of strong nonlinear interactions before applying DASH. The FSI diagnostic (Section 6.4) can help: if FSI values are low across all features, ensemble averaging may not be needed.

7 Discussion

7.1 Reframing explanation instability

This work reframes the SHAP instability problem from "SHAP is noisy under multicollinearity" to a specific mechanistic hypothesis: *sequential residual dependency in gradient boosting creates a first-mover bias that concentrates feature importance on arbitrary features within correlated groups.* Three lines of evidence support this view: (1) the Large Single Model—which maximizes sequential dependency—produces the poorest reproducibility at every correlation level; (2) all methods that achieve model independence restore stability to the same level (≈ 0.977 at $\rho = 0.9$), regardless of their other design choices; and (3) the effect scales with correlation severity, as the hypothesis predicts. The independence principle would be falsified by a setting in which averaging over independently trained models produces *lower* stability than a single model—a scenario we have not observed but cannot rule out under adversarial construction or extreme model heterogeneity (e.g., if independently

trained models learn qualitatively different functions rather than merely different feature orderings).

Isolating independence from averaging. A methodological limitation of our experimental design is that we cannot fully disentangle the effect of *averaging* K estimates from the effect of *independence* among those estimates. Any single XGBoost model's SHAP values are already implicitly aggregated across its constituent trees by TreeSHAP—a 2000-tree model's attributions reflect contributions from all 2000 sequentially dependent trees. Ensemble SHAP (Table 4) effectively "averages" across these dependent trees yet achieves only 0.956 stability, well below the independent-tier methods ($\approx$ 0.977). This between-tier gap provides the strongest available evidence that independence, not merely the number of trees or the act of aggregation, is the operative factor. A colsample ablation (Appendix C, §C) provides direct evidence: DASH and Random Selection with unrestricted `colsample_bytree` (0.5–1.0) achieve only Single-Best-level stability (0.953 vs. 0.958), while the same methods with low `colsample_bytree` (0.1–0.5) achieve 0.976–0.977 ($p < 0.001$). At $\rho = 0$ the colsample range has no effect ($p = 0.54$), confirming that forced feature restriction is specifically needed under collinearity. The one remaining gap is a direct test of Stochastic Retrain with unrestricted `colsample_bytree`; until such evidence is available, our central claim should be read as: model independence *combined with* forced feature restriction is the operative mechanism, with the colsample ablation showing that restriction alone accounts for a +0.023 stability gap and the Ensemble SHAP counter-example showing that aggregation without independence is insufficient.

Positioning relative to prior work. Our contribution is complementary to, not competitive with, the Rashomon-set literature. Fisher et al. [2019] characterize the *range* of variable importance across near-optimal models; we show how to *aggregate* across that range to produce stable point estimates. Bilodeau et al. [2024] prove that no single-model attribution method can reliably identify feature effects across sufficiently rich model classes; our independence principle is a constructive response—by averaging across independently trained models, we circumvent the single-model impossibility. The companion work further quantifies the cost of stability: any stable attribution method is unfaithful to at least one model in every Rashomon pair, implying a worst-case unfaithfulness rate of $\geq$50% for binary sign attribution on Rashomon-connected features. Chen et al. [2024] provide an information-theoretic framework for gradient-boosting Rashomon sets; our first-mover bias hypothesis offers a complementary *mechanistic* account specific to SHAP attributions. In each case, the prior work establishes the *problem* (importance instability is fundamental); we contribute a *resolution* (model-independent averaging) and *diagnostics* (FSI, IS Plot) for detecting when the problem affects a particular analysis.

Quantitative bridge via mutual information. The companion work also establishes a quantitative chain linking feature dependence to explanation error: if two features share positive mutual information ($I > 0$), there exist Rashomon-equivalent models whose attributions for those features have opposite signs; the triangle inequality then implies that any stable explanation has error $\geq \Delta/2$ on at least one witness, where Δ is the attribution gap between the models. This floor is unreachable—no algorithm can beat it. Mutual information thus serves as the *exact boundary* for the impossibility: $I > 0$ triggers it, $I = 0$ does not. Standard correlation-based diagnostics (Pearson ρ, VIF) miss nonlinear dependencies entirely (e.g., $X_2 = X_1^2$ yields $|\rho| \approx 0$, VIF ≈ 1, but $I = 1.91$). On the Breast Cancer dataset, MI-based pre-screening identifies 292 of 435 feature pairs as structurally dependent despite low linear correlation, each subject to the $\Delta/2$ error floor. DASH resolves this by changing the question: instead of producing a single ranking (which must violate either faithfulness or stability when $I > 0$), it aggregates across models to produce a stable

consensus at the cost of decisive pairwise ordering—the enrichment that the companion paper proves is the *only* structural escape from collapsed tightness.

The implication is that the problem is not with SHAP itself—whose axiomatic properties (local accuracy, missingness, consistency) are sound for a given model—but with explaining a single sequentially-constructed model. The solution is to change what is being explained: from one model's feature attributions to a consensus across independently trained models. Under nonlinear data-generating processes, independence remains the most effective mitigation but does not fully restore stability (Section 6.6). The mechanism is that SHAP attributions for a feature depend on *which interactions* the model has learned, not only on which feature was selected first within a correlated group. Since independently trained models may learn qualitatively different interaction structures (e.g., tree h_1 in model A captures z_1^2 while model B captures $z_1 z_2$), averaging their SHAP values introduces model-disagreement noise that is distinct from first-mover bias and not resolved by independence alone.

Random forests as an independence control. Random Forest (RF) provides a natural test of the independence principle from the model-family side: its trees are trained independently by construction (via bootstrap sampling and random feature subsets). The companion impossibility analysis [Caraker et al., 2026b] predicts this theoretically: RF's attribution ratio converges as $O(1/\sqrt{T})$, in contrast to gradient boosting's divergent $1/(1-\rho^2)$—independence between trees prevents the compounding that drives first-mover concentration. In the linear correlation sweep, RF achieves stability competitive with DASH (0.978 vs. 0.977 at $\rho = 0.9$), consistent with this prediction. However, RF's top-5 ranking agreement is substantially lower (0.375 vs. 0.863; $p < 0.001$, $d = +0.488$) and its predictive RMSE is significantly worse (0.957 vs. 0.591). This pattern—high stability but poor top-feature identification—indicates that RF produces *stable but systematically different* attributions. Independence ensures that arbitrary choices cancel upon averaging, yielding high stability; but if the base models are individually inaccurate (e.g., due to high prediction error), the stable consensus may converge to an incorrect ranking. DASH achieves both stability and accurate attributions because its base models are individually performant XGBoost learners filtered for predictive quality before aggregation.

Relationship to stability selection. DASH shares a structural resemblance with stability selection [Meinshausen and Bühlmann, 2010]: both train many models and aggregate their outputs to identify stable signals. The key distinction is the axis of perturbation. Stability selection perturbs the *data* (via subsampling) and aggregates binary feature *selection* indicators, identifying which features are consistently chosen across subsamples. DASH perturbs the *model* (via hyperparameter diversification) and aggregates continuous feature *attributions* (SHAP values), producing a stable importance ranking over the original feature space. The two approaches are complementary: stability selection operates at the feature-selection stage and discards correlated redundancies, while DASH operates at the explanation stage and preserves all features, distributing credit proportionally within correlated groups. A practitioner who has already applied stability selection to reduce the feature set would still benefit from DASH when explaining the model trained on the selected features, as within-group collinearity among retained features can persist.

The DASH ensemble's K importance vectors also support a richer output representation: features that consistently swap rank across ensemble members can be treated as *incomparable* rather than arbitrarily ordered, yielding a partial order on feature importance that more honestly reflects genuine attribution ambiguity under collinearity. We pursue this direction in companion work [Caraker et al., 2026b], which formalizes the conditions under which partial orders are necessary

rather than merely convenient.

7.2 Practical recommendations

1. **Always check for first-mover bias.** Before trusting a SHAP-based feature ranking from a gradient-boosted model, train multiple models with different seeds and compare their importance rankings. If rankings differ substantially, the standard workflow is unreliable.

2. **Use DASH's diagnostics.** The FSI and IS Plot detect which specific features are affected by first-mover bias, even without ground truth. Quadrant II features (high importance, high instability) should be interpreted as *collinear cluster members* rather than individually important features.

3. **For stability alone, seed averaging suffices.** Stochastic Retrain achieves stability equivalent to DASH with minimal implementation effort. Use it when diagnostics and equity are not required.

4. **Use DASH when equity and diagnostics matter.** DASH's forced feature restriction produces lower within-group CV, and its integrated diagnostics provide actionable audit information.

5. **Do not scale up single models.** The Large Single Model result demonstrates that more trees with sequential training amplifies, rather than resolves, explanation instability.

When to use which method. As a rough heuristic: (1) if feature correlations are low ($\rho < 0.5$) and the dataset is moderate-sized, a single well-tuned model's SHAP values are likely adequate—our results show negligible stability differences in this regime; (2) if moderate correlations exist ($0.5 \leq \rho < 0.8$) and stability is the primary concern, Stochastic Retrain (seed averaging) provides the simplest solution with minimal implementation effort; (3) if correlations are high ($\rho \geq 0.8$), equitable credit distribution within correlated groups matters, or ground-truth-free diagnostics are needed, use the full DASH pipeline. In all cases, the FSI and IS Plot can serve as a preliminary collinearity audit regardless of which explanation method is ultimately adopted.

Memory and compute. DASH's peak memory footprint scales as $O(M \times S_{\text{model}})$ during the population stage, where S_{model} is the in-memory size of a single XGBoost model (typically 1–10 MB for models with moderate depth and early stopping). At $M = 200$ this can require 200 MB–2 GB for the model population; the SHAP computation stage adds $O(K \times N' \times P)$ for the attribution matrices. For wall-clock timing, see Table 11.

Model governance considerations. In regulated settings (e.g., banking model risk management under SR 11-7, or transparency obligations under the EU AI Act), DASH is best understood as an *explanation auditing layer* rather than a replacement for the production model. The deployed system still trains and serves a single XGBoost model; DASH runs offline during model validation to assess whether that model's SHAP-based feature rankings are trustworthy. The M-model population is an audit instrument— analogous to stress-testing a model under varied specifications—not a deployed ensemble. For model risk documentation, we recommend recording: (1) the production model's SHAP rankings alongside the DASH consensus rankings, (2) the FSI values and IS Plot quadrant assignments as evidence of explanation reliability or instability, and (3) specific features flagged as Quadrant II members, which should be reported as collinear groups rather than individually ranked.

DASH itself need not be separately inventoried as a model; it is a diagnostic tool applied to an existing inventoried model, analogous to backtesting or stress testing.

Expected improvement magnitude. The Breast Cancer result (+0.549 over Single Best) reflects extreme collinearity (21 feature pairs with $|r| > 0.9$). On datasets with moderate collinearity—California Housing ($|r| \approx 0.7$, 8 features)—the stability improvement is +0.009 (not significant), and DASH's primary value is the diagnostic output rather than the stability gain. Practitioners should calibrate expectations accordingly: improvements of +0.01–0.05 on stability are typical for moderately correlated datasets, with larger gains reserved for settings with heavy collinearity.

7.3 Limitations

- **Scope of the independence principle.** The empirical validation in this paper covers interventional TreeSHAP on XGBoost. The companion impossibility theorem [Caraker et al., 2026b] proves that the underlying instability—and the necessity of ensemble-based resolution—holds for *any* attribution method on *any* model class exhibiting the Rashomon property under correlation, providing a theoretical basis for expecting the independence principle to generalize. However, direct empirical validation on other configurations (conditional SHAP, LightGBM, CatBoost, neural networks with KernelSHAP) remains future work. The effect pattern may differ for alternative GBDT implementations: CatBoost's ordered boosting randomizes the training-example ordering rather than the feature ordering, potentially reducing first-mover bias through a different mechanism; LightGBM's histogram-based leaf-wise growth may concentrate importance differently than XGBoost's level-wise strategy. The `fit_from_attributions()` interface partially bridges this gap: any attribution method producing feature-level vectors can plug into DASH's aggregation stages without retraining (validated on LIME in Appendix G). The conditional SHAP case warrants particular attention: although conditional approaches account for feature dependencies in the reference distribution [Aas et al., 2021], the companion paper proves that switching to conditional SHAP does *not* escape the impossibility when features have equal causal effects [Caraker et al., 2026b]—the instability is a property of the Rashomon set, not the attribution method. However, the practical dynamics of instability under conditional SHAP may differ from the interventional setting, and empirical validation remains open.

- **Interventional SHAP under correlation.** DASH uses interventional TreeSHAP, which conditions on marginal rather than conditional feature distributions. Under high correlation, this evaluates the model at out-of-distribution feature combinations [Janzing et al., 2020, Aas et al., 2021], potentially producing unintuitive attributions. Averaging across diverse models mitigates individual-model artifacts, but the fundamental tension between interventional SHAP and correlated features remains. More broadly, DASH stabilizes *predictive* attributions—what the model learned from each feature—not *causal* attributions in the interventionist sense [Janzing et al., 2020]. The term "causally active" used in Appendix E refers to features with non-zero coefficients in the data-generating process, not to a formal causal framework. The semantics of interventional SHAP under collinearity are themselves disputed [Chen et al., 2020, Aas et al., 2021]; DASH does not resolve this debate but reduces the *additional* instability introduced by model-training randomness.

- **Interaction effects.** The current pipeline averages SHAP value matrices $\Phi^{(i)} \in \mathbb{R}^{N' \times P}$, which preserves main effects but not pairwise interaction structure. However, TreeSHAP supports exact interaction values via tensors $\Phi_{\text{int}}^{(i)} \in \mathbb{R}^{N' \times P \times P}$, where diagonal entries are main effects

and off-diagonal entries are pairwise interactions. Averaging these tensors element-wise across the ensemble would yield stable interaction estimates by the same independence argument. The computational cost is $O(TLD^2)$ per model (vs. $O(TLD)$ for standard SHAP), making this practical for moderate P but expensive for large feature spaces.

- **Nonlinear scope boundary.** Under nonlinear DGPs, overall stability is lower for all methods. DASH's advantage grows with ρ and is clearest at $\rho \geq 0.7$ (Section 6.6).

- **Ground truth.** On real-world data, we can only evaluate stability, not accuracy. The synthetic accuracy metric presupposes equitable credit distribution (Section 5), partially confounding accuracy with equity. Specifically, the linear DGP assigns equal true importance to all features within a correlated group by construction, aligning the equity and accuracy metrics by design rather than testing them independently. Appendix E examines the asymmetric case where one feature is causally active and its correlate is a passive proxy, directly testing whether DASH over-equalizes.

- **Background dataset size.** All experiments use $B = 100$ background samples for interventional TreeSHAP. For high-dimensional datasets with strong correlation structure, this may be insufficient to capture the joint distribution faithfully.

- **Column ordering.** XGBoost's split selection can be influenced by the order in which features appear in the data matrix when candidate splits have equal gain. We have not tested whether randomly permuting column order across repetitions affects the stability results. Because DASH trains models with random `colsample_bytree` subsets, column-order effects are partially mitigated by the stochastic feature sampling, but a systematic robustness test remains future work.

- **Sign stability.** This paper evaluates *ranking* stability (Spearman correlation of importance magnitudes) but does not assess *sign* stability—whether a feature's SHAP contribution is consistently positive or negative across retraining runs. The companion impossibility theorem [Caraker et al., 2026b] proves a bilemma showing that sign stability and faithfulness are jointly impossible for features in Rashomon pairs: no method can consistently assign the correct sign across equivalent models. DASH consensus averaging resolves ranking instability (via the neutral element of tied ranks) but cannot resolve sign instability (binary signs admit no neutral element). Empirical characterization of sign flip rates under DASH is deferred to future work.

7.4 Broader implications

First-mover bias is likely not unique to gradient boosting. Any iterative optimization procedure that makes sequential, greedy feature selections—including some neural network training dynamics—may exhibit analogous path-dependent concentration of feature attributions. The independence principle established here provides a general framework for investigating and resolving such effects. Our Random Forest results confirm that tree-level independence already yields high stability (Section 7), validating the principle on a model family that is independent by construction. Future work will explore whether the mechanism extends to neural networks (where gradient-based optimization creates different but potentially analogous path dependencies) and whether partial orders on feature importance can replace point rankings as a more robust representation of explanation structure.

The empirical equivalence between DASH and Stochastic Retrain—and the monotonic degradation of dependent methods with ρ—is consistent with a formal impossibility result we establish

in companion work [Caraker et al., 2026b]: no single-model feature ranking can simultaneously be faithful (reflect the model's attributions), stable (consistent across equivalent models), and complete (rank all feature pairs) when features are collinear. This trilemma holds for any attribution method applied to any model class exhibiting the Rashomon property under correlation—not only gradient boosting. The present work provides the empirical foundation and constructive resolution; the companion theorem, mechanically verified in the Lean 4 proof assistant, proves that ensemble-based approaches like DASH are not merely effective but mathematically necessary for achieving both stability and equity.

A stronger result also holds: the companion paper proves a *bilemma* showing that even without the completeness requirement, no method for determining a feature's SHAP sign (positive vs. negative contribution) can be simultaneously faithful and stable across Rashomon-equivalent models. The bilemma is constructive and zero-axiom in Lean 4, requiring only the existence of two models with identical predictions but opposite SHAP signs—a condition satisfied by construction in any correlated feature group. A corollary (the all-or-nothing theorem) predicts that SHAP sign flip rates should exhibit *bimodality*: features either flip near 0% (no Rashomon ambiguity) or $\geq$50% (full Rashomon ambiguity), with a dead zone in between. This prediction is directly testable against the existing experimental data and constitutes a falsifiable consequence of the theoretical framework. DASH resolves *ranking* instability because feature rankings admit a neutral element (tied rank under consensus averaging), but the bilemma proves that *sign* instability is irreducible: binary signs have no neutral element, so no averaging scheme can simultaneously preserve faithfulness and achieve stability for sign attribution.

The neural network case warrants particular attention. Attribution instability under feature collinearity is well-documented for NNs: small input perturbations produce large ranking changes [Ghorbani et al., 2019], and independently trained models achieving indistinguishable test loss can produce divergent Shapley rankings [D'Amour et al., 2020]—a direct analog of GBDT first-mover bias operating through initialization-driven symmetry breaking rather than residual sequencing. The Rashomon set framework [Fisher et al., 2019, Semenova et al., 2022] predicts that averaging over independent initializations should cancel this bias by the same argument that motivates DASH: the ensemble mean converges toward the attribution implied by the basin center of mass, not any single sample's idiosyncratic credit assignment. However, two differences temper this prediction. First, NNs possess a wider Rashomon set under high collinearity [D'Amour et al., 2020], likely requiring larger K for equivalent stability gains. Second, gradient-based attribution methods (Integrated Gradients, GradSHAP) introduce baseline-sensitivity variance absent in TreeSHAP [Alvarez-Melis and Jaakkola, 2018], confounding attribution instability with method instability in any NN experiment. A careful extension of DASH to neural architectures would need to control for both factors—a promising but non-trivial direction. A concrete near-term extension is stable feature interaction estimation: averaging TreeSHAP interaction tensors across the DASH ensemble would provide stable pairwise interaction rankings under multicollinearity, a capability that no single-model workflow can offer.

8 Conclusion

We have investigated first-mover bias—the path-dependent concentration of feature importance associated with sequential residual fitting in gradient boosting—as a specific mechanistic contributor to SHAP instability under multicollinearity. Three lines of evidence support this finding:

1. The Large Single Model, which maximizes sequential dependency, produces the poorest attribution reproducibility of any method tested—worse than the standard single-best workflow—

despite matching DASH's total tree count and feature restriction. This provides strong evidence that sequential dependency, not model capacity, is a major driver of instability.

2. DASH and Stochastic Retrain achieve equivalent stability (0.977 at $\rho = 0.9$; accuracy $d = +0.05$, equity $d = -0.13$, both n.s.; TOST confirmed), despite differing in every design choice except model independence. This supports the view that independence between explained models is the operative mechanism. DASH's diversity-aware selection achieves this with $K_{\text{eff}} \approx 12$ models (vs. $K = 30$ for SR) and provides significantly more equitable attribution spreading within correlated groups ($p < 0.001$ vs. Random Selection at every ρ level).

3. The effect scales with correlation: dependent methods degrade monotonically as ρ increases, while independent methods remain flat. This matches the mechanistic prediction.

DASH (Diversified Aggregation for Stable Hypotheses) operationalizes the independence principle at two layers: the *core operation*—averaging attributions across independent models, proved to be the minimum-variance unbiased estimator [Caraker et al., 2026b]—and the *pipeline*, which adds forced feature restriction, diversity-aware model selection, and two diagnostic tools (the Feature Stability Index and Importance-Stability Plot) that detect first-mover bias without ground truth. On real-world datasets, DASH improves stability from 0.339 to 0.925 (Breast Cancer), from 0.840 to 0.964 (Superconductor), and from 0.97 to 0.98 (California Housing).

The code, data, and reproducible benchmarks are publicly available at https://github.com/DrakeCaraker/dash-shap.

Reproducibility Statement

All code, data generators, hyperparameter configurations, and evaluation metrics are publicly available at https://github.com/DrakeCaraker/dash-shap. The complete benchmark can be reproduced via `python run_experiments_parallel.py` with fixed random seeds (SEED = 42). The authoritative interactive notebook `notebooks/demo_benchmark_7_parallel.ipynb` provides a checkpointed walkthrough of all experiments. The canonical results were produced on AWS SageMaker (`ml.g5.16xlarge`, 64 vCPU, 248 GB RAM); hardware-dependent timing results (Table 11) were measured on Apple M-series silicon, single node, and are reported for relative comparison only. Key software versions for the canonical results: Python 3.11.11, XGBoost 2.1.4, SHAP 0.51.0, NumPy 2.3.5, SciPy 1.14.1, scikit-learn 1.6.1, joblib 1.4.2. Full environment specifications (including BLAS/LAPACK configuration) are recorded in `results/environment.json`. We note that floating-point non-determinism across hardware platforms (e.g., SageMaker Intel vs. Apple M-series ARM) can produce small numerical differences in SHAP values at the least-significant digits; our stability conclusions are robust to these differences because they are based on rank correlations, which are invariant to monotone transformations of the SHAP magnitudes.

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A Algorithm Pseudocode

B Extended Results and Reproducibility

Full per-repetition importance vectors, significance test results, and ablation data are available in JSON format from the experimental runner. Running `python run_experiments_parallel.py` reproduces all tables and figures. The authoritative notebook `notebooks/demo_benchmark_7_parallel.ipynb` provides a checkpointed walkthrough of all experiments with intermediate results cached for rapid re-execution.

All code, data generators, and benchmark infrastructure are publicly available at https://github.com/DrakeCaraker/dash-shap.

C Ablation Studies and Computational Cost

Epsilon sensitivity. DASH is remarkably robust to the performance filter threshold ε (Table 10). Across a $3\times$ range of ε values (0.03 to 0.10), stability varies by < 0.005. The effective ensemble size K_{eff} scales with ε (4.0 to 16.2), but performance plateaus early.

Population size ablation. Stability is robust across population sizes M: $M = 50$ (0.9728) $\rightarrow$ $M = 100$ (0.9759) $\rightarrow$ $M = 200$ (0.9767) $\rightarrow$ $M = 500$ (0.9777). Performance is effectively invariant to population size (within 0.001 across $M \in \{50, 100, 200, 500\}$). We use $M = 200$ as a conservative default.

Algorithm 1 DASH Pipeline

Require: Training data $(X_{\text{train}}, y_{\text{train}})$, validation data $(X_{\text{val}}, y_{\text{val}})$, reference data X_{ref}, population size M, max ensemble size K, performance threshold ε, diversity threshold δ, search space Θ

Ensure: Consensus SHAP matrix $\bar{\Phi}$, diagnostics (FSI, IS Plot)

Stage 1: Population Generation
for $i = 1$ to M **do**
  Sample $\theta_i \sim \text{Uniform}(\Theta)$
  Train $f_i \leftarrow \text{XGBoost}(X_{\text{train}}, y_{\text{train}}; \theta_i, \text{seed} = i)$
  Evaluate $s_i \leftarrow \text{score}(f_i, X_{\text{val}}, y_{\text{val}})$
end for
Stage 2: Performance Filtering
$s^* \leftarrow \max_i s_i$
$\mathcal{F} \leftarrow \{i : |s_i - s^*| \leq \varepsilon\}$
Stage 3: Diversity Selection
for $i \in \mathcal{F}$ **do**
  $\mathbf{v}_i \leftarrow \text{gain_importance}(f_i, X_{\text{ref}})$
end for
$\mathcal{S} \leftarrow \text{MaxMinSelect}(\{\mathbf{v}_i\}_{i \in \mathcal{F}}, K, \delta)$
Stage 4: Consensus SHAP
for $i \in \mathcal{S}$ **do**
  $\Phi^{(i)} \leftarrow \text{TreeSHAP}(f_i, X_{\text{ref}})$
end for
$\bar{\Phi} \leftarrow \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \Phi^{(i)}$
Stage 5: Diagnostics
$\bar{I}_j \leftarrow \frac{1}{N'} \sum_n |\bar{\Phi}_{nj}|$ for each feature j
$\text{FSI}_j \leftarrow \bar{\sigma}_j / (\bar{I}_j + \epsilon_0)$ for each feature j
return $\bar{\Phi}$, FSI, IS Plot

Table 10: Epsilon sensitivity at $\rho = 0.9$ (50 repetitions). Performance is robust across a 3× range.

ε	Models Passing	K_{eff}	Stability	Accuracy	Equity
0.03	9.3	4.0 ± 1.4	0.9759	0.9876	0.183
0.05	21.0	6.6 ± 1.9	0.9764	0.9879	0.178
0.08	46.1	11.9 ± 3.0	0.9767	0.9881	0.175
0.10	63.0	16.1 ± 3.6	0.9774	0.9884	0.173

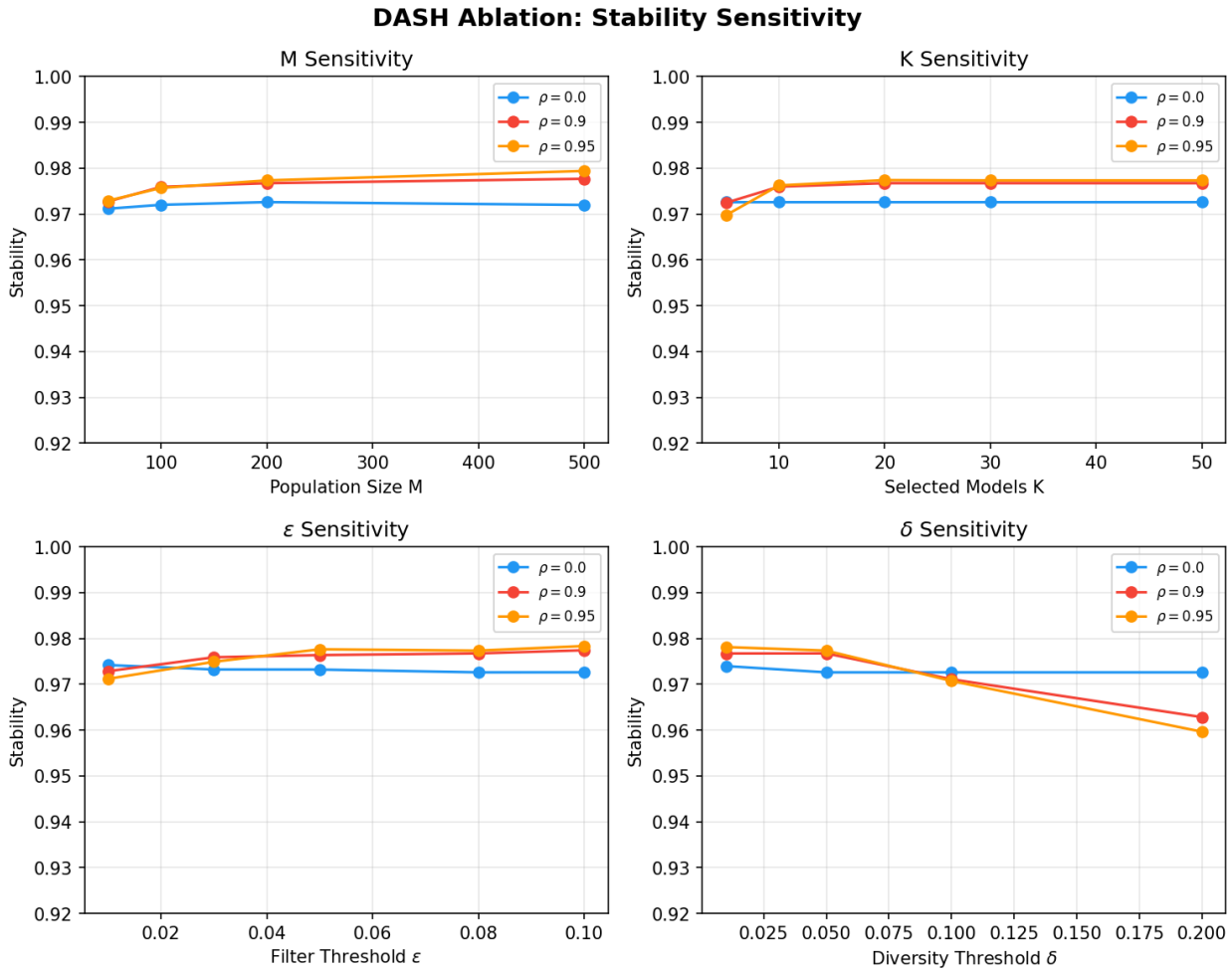

Figure 6: Ablation sensitivity: stability as a function of each DASH hyperparameter at three correlation levels ($\rho \in \{0.0, 0.9, 0.95\}$). **Top left**: Population size M—stability is effectively invariant ($\Delta < 0.005$ across $M \in \{50, 100, 200, 500\}$). **Top right**: Selected models K—saturates at $K \approx 20$. **Bottom left**: Filter threshold ε—robust across a $3\times$ range (0.03–0.10). **Bottom right**: Diversity threshold δ—sensitive above 0.05; the default $\delta = 0.05$ is at the elbow.

Colsample restriction ablation. To isolate whether forced low `colsample_bytree` is the operative diversity mechanism, we train DASH and Random Selection under three `colsample_bytree` ranges: Low (0.1–0.5, the default), High (0.5–1.0), and Full (0.1–1.0), with Stochastic Retrain and Single Best as controls.

At $\rho = 0$ (safety control), the colsample range has no effect: DASH Low and High achieve identical stability (0.972 vs. 0.973; $p = 0.54$). At $\rho = 0.9$, low colsample significantly outperforms high colsample for both DASH (0.976 vs. 0.953; $\Delta = +0.023$, $p < 0.001$, bootstrap CIs non-overlapping) and Random Selection (0.977 vs. 0.957; $\Delta = +0.020$, $p < 0.001$). Notably, DASH with high `colsample_bytree` (0.953) performs comparably to Single Best (0.958), indicating that ensemble averaging *without* forced feature restriction does not recover DASH-level stability. Under nonlinearity at $\rho = 0.9$, the gap widens: DASH Low achieves 0.885 vs. High 0.801 ($\Delta = +0.084$; bootstrap CIs non-overlapping).

Computational cost. Table 11 reports wall-clock time for each method at $\rho = 0.9$ (50 repetitions, single-threaded timing). DASH's cost is dominated by training $M = 200$ models and computing $K \leq 30$ TreeSHAP explanations.

Table 11: Computational cost at $\rho = 0.9$ (50 repetitions). Wall-clock times are hardware-dependent (Apple M-series, single thread) and reported for relative comparison; absolute times on the SageMaker `ml.g5.16xlarge` instance are $\sim$12$\times$ higher due to `nthread=1` per-rep parallelism.

Method	**Models**	**SHAP Evals**	**Per-rep (s)**	**Ratio**
Large Single Model	1	1	6.4	0.1$\times$
Single Best	30	1	43.5	1.0$\times$
LSM (Tuned)	1	1	88.9	2.0$\times$
DASH (MaxMin)	200	K_{eff}	140.3	3.2$\times$
Stochastic Retrain	30	30	233.5	5.4$\times$
Single Best (M=200)	200	1	248.8	5.7$\times$
Random Selection	200	K_{eff}	287.2	6.6$\times$

Note: DASH's diversity selection typically terminates before reaching $K_{\max} = 30$ (the minimum-distance threshold δ stops selection early), yielding $K_{\text{eff}} \approx 10$–15 SHAP evaluations at $\varepsilon = 0.08$. Random Selection always selects $K = 30$ models, requiring roughly twice as many SHAP computations per repetition. Stochastic Retrain similarly computes SHAP for all $K = 30$ models, explaining its higher cost relative to DASH.

D Pre-Specified Success Criteria

We pre-specified eleven pass/fail criteria before running the final benchmark (written into the experimental notebook prior to execution, though not lodged with a formal pre-registration registry). These criteria test the paper's stated hypotheses under favorable conditions (known-DGP synthetic data and datasets with documented collinearity); adversarial or out-of-distribution stress tests are deferred to the journal version. DASH passes all eleven:

1. **Stability wins (linear)**: DASH > Single Best on $\geq$4/5 ρ levels $\rightarrow$ **PASS** (4/5)
2. **Accuracy at** $\rho = 0.9$: DASH $\geq$ SB $\rightarrow$ **PASS** (DASH = 0.9881 vs. SB = 0.9782)
3. **Equity wins (linear)**: DASH < Single Best CV on $\geq$4/5 ρ levels $\rightarrow$ **PASS** (5/5)
4. **Safety control at** $\rho = 0$: No degradation vs. baselines $\rightarrow$ **PASS** (gap = 0.0003)
5. K_{eff} **increases with** ε: monotonic $\rightarrow$ **PASS** (4.0 $\rightarrow$ 6.6 $\rightarrow$ 11.9 $\rightarrow$ 16.1)
6. **Nonlinear DGP**: DASH > SB stability at $\rho = 0.9$ $\rightarrow$ **PASS** (DASH = 0.887 vs. SB = 0.811)
7. **Statistical significance**: $\geq$50% of tests significant $\rightarrow$ **PASS** (17/26 Bonferroni, 15/26 Holm–Bonferroni)
8. **Superconductor**: DASH stability > SB $\rightarrow$ **PASS** (0.964 vs. 0.840)
9. **California Housing**: DASH stability > SB $\rightarrow$ **PASS** (0.978 vs. 0.969; $p = 0.063$ n.s. but positive direction)
10. **Breast Cancer**: DASH stability > SB $\rightarrow$ **PASS** (0.925 vs. 0.376)
11. **Variance decomposition**: DASH model-selection variance < SB $\rightarrow$ **PASS** (0.006 vs. 0.023)

Self-critique. All eleven criteria pass, which may suggest that the criteria were not sufficiently stringent to risk failure on informative dimensions. The most revealing potential criterion would have been "DASH stability > SR stability (statistically significant)," which our data does *not* support—SR matches or marginally exceeds DASH on stability at every ρ level. We did not pre-specify this as a success criterion because our hypothesis is that *independence* drives stability (which SR also achieves), not that DASH's specific pipeline outperforms SR. Had we included "DASH > SR on stability," it would have failed—and this failure is itself informative, as it constrains the contribution of MaxMin diversity selection to equity and diagnostics rather than stability. We encourage readers to treat the SR equivalence as the most informative null result in this paper.

E Asymmetric Causal DGP Check

Motivation. The linear DGP used in the main experiments assigns equal true importance to all features within a correlated group, which means that DASH's equity metric (within-group CV) is evaluated in a regime where equity and accuracy are perfectly aligned by construction. A natural concern is whether DASH *over-equalizes* when one feature is causally active and its correlate is a passive proxy—that is, does aggregation over independent models wash out a legitimate importance asymmetry?

DGP. We use a two-feature asymmetric causal DGP:

$$y = 2f_0 + \varepsilon, \quad f_1 = \rho f_0 + \sqrt{1-\rho^2}\, z, \quad \varepsilon, z \sim \mathcal{N}(0, \sigma^2),$$

where f_0 is causally active and f_1 is a passive correlate. Ground-truth importance: $f_0 = 1.0$, $f_1 = 0.0$ (normalized). We sweep $\rho \in \{0.5, 0.7, 0.9, 0.95\}$ with $N = 5{,}000$ and $N_{\text{reps}} = 50$ repetitions.

Metrics. For each method we report:

- **Stability**: mean pairwise Spearman correlation across repetitions (same as the main experiments).
- **Attribution bias**: $|\overline{\hat{\phi}}_0 - \text{true}(f_0)|$, the absolute deviation of the mean SHAP attribution for f_0 from its ground-truth importance (after normalizing attributions to sum to 1).
- **Passive leak**: mean attribution assigned to f_1 (ground truth: 0.0); a proxy for over-equalization.

Results. Table 12 reports results at $\rho = 0.9$ (the primary evaluation condition in the main experiments). DASH correctly preserves the causal asymmetry: all methods attribute substantially more importance to the causal feature f_0 than the passive correlate f_1. However, DASH's passive leak (0.089) is higher than Single Best's (0.068), reflecting a trade-off inherent to ensemble averaging: some models in the ensemble attribute to f_1, and their contributions are not zeroed out by averaging. Stochastic Retrain shows a similar pattern (0.074). This trade-off increases with ρ (DASH passive leak rises from 0.046 at ρ=0.5 to 0.173 at ρ=0.95), and is the expected cost of the variance reduction that DASH provides.

Table 12: Asymmetric causal DGP at $\rho = 0.9$, 50 repetitions. "Passive leak" is mean attribution to the causally inert feature f_1 (ground truth: 0.0). Lower is better for both bias and passive leak. All methods achieve stability = 1.000 (deterministic DGP).

Method	**Stability**	**Bias (f_0)**	**Passive leak (f_1)**
Single Best	1.000	0.068	0.068
Stochastic Retrain	1.000	0.074	0.074
DASH (MaxMin)	1.000	0.089	0.089
Large Single Model	1.000	0.084	0.084

F Two-Tree Analytical Example

To build intuition for first-mover bias, we analyze the simplest possible case: two-step linear boosting on correlated features. The companion paper [Caraker et al., 2026b] proves the formal result—the attribution ratio is exactly $1/(1-\rho^2)$ for gradient boosting under the proportionality axiom—and verifies it in Lean 4. Here we provide an accessible derivation that isolates the core mechanism.

Setup. Let A, B be two features with $\mathrm{Cov}(A, B) = \rho$, $\mathrm{Var}(A) = \mathrm{Var}(B) = 1$, and target $y = A + \varepsilon$. Consider linear boosting with learning rate η: at each step we fit y (or its residual) by regressing on a single feature and updating the model with the fitted value scaled by η.

Step 1. Suppose the first step selects feature A. The fitted coefficient is $\hat{\beta}_1 = \mathrm{Cov}(y, A)/\mathrm{Var}(A) = 1$, so the model after step 1 is $f_1(x) = \eta A$. The residual is $r_1 = y - \eta A = (1-\eta)A + \varepsilon$.

Step 2. Now consider the gain from each feature on the residual r_1:

$$\mathrm{Gain}(A) = \big[\mathrm{Cov}(r_1, A)\big]^2/\mathrm{Var}(A) = (1-\eta)^2, \tag{10}$$

$$\mathrm{Gain}(B) = \big[\mathrm{Cov}(r_1, B)\big]^2/\mathrm{Var}(B) = \rho^2(1-\eta)^2. \tag{11}$$

Since $\rho < 1$, feature A has strictly higher gain on the residual: $\mathrm{Gain}(A)/\mathrm{Gain}(B) = 1/\rho^2$. The second step selects A again. After two steps, the model is $f_2(x) = \eta(2-\eta)A$: *only feature A is used*, and all SHAP credit goes to A.

From gain bias to concentration. The gain ratio $1/\rho^2 \approx 1.23$ at $\rho = 0.9$ demonstrates the mechanism: after one step, the residual structure encodes which feature was used, creating a bias toward re-selecting that feature.

This simple linear model is deliberately minimal. Two features and linear gain are sufficient to isolate the residual-bias mechanism, but they do not capture the full concentration dynamics of tree-based boosting. In particular, when the DGP is symmetric ($y = \beta\bar{z}_g + \varepsilon$ with $\bar{z}_g = (A+B)/2$, the paper's main setting), linear boosting alternates between A and B at each step—the unused feature always has marginally higher gain—so no net concentration arises in this idealization.

In actual XGBoost, three additional factors break this alternation and enable the concentration observed in Figure 1: (1) threshold-based splits create nonlinear gain functions that do not alternate as cleanly as linear regression gains; (2) multi-depth trees create interaction structure where a feature used at the root influences which features are useful at child nodes; and (3) `colsample_bytree` restricts the available feature set per tree, so the feature that accumulates slightly more early splits

(by chance) builds a residual advantage that compounds over hundreds of trees. The interplay of these stochastic and structural effects is what produces the monotonically growing concentration in Figure 1—the linear gain-bias derived above provides the seed, and tree-specific dynamics amplify it.

Limitations of this analytical model. This two-feature linear model demonstrates the *seed mechanism* (residual gain bias of $1/\rho^2$) but does not constitute a proof that concentration occurs in full XGBoost under the paper's symmetric DGP. The bridge from the analytical seed to the observed concentration is empirical: Figure 1 shows monotonically growing concentration as tree count increases in actual XGBoost, which is consistent with the seed mechanism being amplified by the three factors above. A complete analytical model of concentration dynamics in multi-depth, stochastic-feature-subset boosting—including the interaction between `colsample_bytree`, multi-level splits, and residual evolution—remains an open theoretical problem.

Independence resolves the compounding. Because independently trained models make their first-mover selections independently, the compounding runs in different directions for different models. When their SHAP vectors are averaged:

$$\bar{\phi}_A \approx \tfrac{1}{K} \sum_{k=1}^{K} \phi_A^{(k)}, \tag{12}$$

models that concentrated on A and models that concentrated on B cancel each other's arbitrary choices. With K independent models, the variance of the consensus importance due to first-mover effects decays as $O(1/K)$, explaining the stability plateau at $K \approx 20$ observed in the K-sweep experiment.

This example identifies the residual gain-bias seed ($1/\rho^2$) and the cancellation principle ($O(1/K)$ variance decay). Figure 1 confirms the empirical predictions: (1) concentration grows monotonically with T for a single sequential model, and (2) independent ensembles remain flat regardless of per-model tree count.

G Attribution-Agnostic Interface Validation

The `fit_from_attributions()` interface (contribution 5 in Section 1) decouples DASH's aggregation stages (filtering, diversity selection, consensus, diagnostics) from the attribution method. We validate this claim by applying DASH to LIME [Ribeiro et al., 2016] attributions on the Breast Cancer dataset.

Procedure. We train $M = 30$ XGBoost classifiers with randomly sampled hyperparameters (same search space as the main experiments, `colsample_bytree` $\in \{0.2, 0.3, 0.4, 0.5\}$). For each model, we compute LIME attributions for $N' = 30$ observations from the held-out explain set, producing an (M, N', P) attribution tensor. We then call `pipe.fit_from_attributions(attributions, val_scores)` with $K = 10$ and relative $\varepsilon = 0.05$.

Results. DASH successfully executes all four post-population stages on the LIME tensor. The consensus top features—`worst concave points` (0.104), `worst texture` (0.102), `worst area` (0.086), `worst perimeter` (0.073)—draw from the same clinically relevant feature families (concavity, area, perimeter) as the TreeSHAP-based rankings in Table 8, confirming that the diagnostic framework

generalizes across attribution methods. MaxMin diversity selection yields $K_{\text{eff}} = 3$ (low because all 30 models achieve identical validation accuracy on this dataset, limiting diversity in the filtered pool).

This demonstrates that DASH's stages 2–5 can operate on any (M, N', P) attribution tensor, regardless of whether it was produced by TreeSHAP, LIME, Integrated Gradients, or any other feature-level attribution method. The independence principle—that averaging over independently trained models cancels arbitrary attribution choices—applies to any attribution method that is sensitive to model specification, not only TreeSHAP.

Limitations of this validation. This demonstration uses only $M = 30$ models (vs. $M = 200$ in the main experiments), yields $K_{\text{eff}} = 3$ due to low population diversity under identical validation accuracy, and evaluates a single dataset. It confirms that the pipeline *executes* on non-TreeSHAP attributions and produces plausible results, but does not constitute a rigorous evaluation of DASH's stability properties under LIME. A full evaluation—with $M = 200$, multiple datasets, and comparison against LIME-specific baselines—is needed before claiming that DASH generalizes to non-TreeSHAP attribution methods.